UNITED STATES
SECURITIES AND EXCHANGE COMMISSION

Washington, DC 20549

FORM 10-Q

(Mark One)

☒ **QUARTERLY REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**
For the quarterly period ended March 31, 2026

OR

☐ **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**
For the transition period from _ to _
Commission File Number: 001-38753



Moderna, Inc.

(Exact Name of Registrant as Specified in Its Charter)

Delaware	**81-3467528**
(State or Other Jurisdiction of Incorporation or Organization)	(IRS Employer Identification No.)
325 Binney Street **Cambridge, Massachusetts**	**02142**
(Address of Principal Executive Offices)	(Zip Code)

(617) 714-6500
(Registrant's Telephone Number, Including Area Code)

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading symbol(s)	Name of each exchange on which registered
Common stock, par value $0.0001 per share	MRNA	The Nasdaq Stock Market LLC

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. **Yes** ☒ **No** ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§ 232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). **Yes** ☒ **No** ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer", "accelerated filer", "smaller reporting company", and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large accelerated filer ☒ Accelerated filer ☐ Non-accelerated filer ☐ Smaller reporting company ☐
Emerging growth company ☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Act). **Yes** ☐ **No** ☒

As of April 24, 2026, there were 396,786,259 shares of the registrant's common stock, par value $0.0001 per share, outstanding.

SPECIAL NOTE REGARDING FORWARD-LOOKING STATEMENTS

This Quarterly Report on Form 10-Q (Form 10-Q) contains express or implied forward-looking statements. All statements other than those of historical facts contained in this Form 10-Q are based on our management's beliefs and assumptions and on information currently available to our management. Forward-looking statements in this Form 10-Q include, but are not limited to, statements about:

- our ability to drive use of our commercial products and expectations regarding a return to sales growth in 2026;
- our ability to execute on our prioritized research and development portfolio;
- our ability to deliver cost efficiency across our business;
- our expectations regarding the size and durability of the respiratory vaccine market;
- the availability of mCOMBRIAX across the EU, subject to national regulatory and access procedures;
- our long-term strategic partnerships with government entities;
- anticipated near-term regulatory actions for our product candidates, including potential filings, reviews and approvals;
- anticipated milestones for our pipeline programs, including potential data readouts and other near-term catalysts;
- our ability to obtain and maintain regulatory approval of our product candidates across our portfolio;
- our ability to successfully launch and commercialize our products and the timing of launches;
- the potential of our oncology portfolio;
- our ability and the ability of third parties with whom we contract to successfully manufacture, supply and distribute our commercial products and any future commercial products at scale, as well as drug substances, delivery vehicles, development candidates, and investigational medicines for preclinical and clinical use;
- financing and funding options we may consider as part of our research and development strategy;
- our ability to successfully contract with third-party suppliers, distributors and manufacturers;
- internal and external costs associated with manufacturing our products and the impact on our cost of sales, and our anticipated cost of sales as a percentage of net product sales;
- the scope of protection we are able to establish and maintain for intellectual property rights, including those covering our commercial products, product candidates and technology, and our expectations regarding pending legal proceedings related to our intellectual property;
- the timing of initiation, progress, completion, results and cost of our clinical trials, preclinical studies and research and development programs, as well as those of our collaborators;
- participant enrollment in our clinical trials, including timing;
- potential advantages of mRNA as compared to traditional medicine;
- the implementation of our business model and strategic plans for our business, products, product candidates and technology;
- the pricing and reimbursement of our products, if approved;
- the build out of our manufacturing and commercial operations;
- estimates of our future expenses, revenues and capital requirements;

- our operation and funding requirements, including our forecast of the period of time through which our financial resources will be adequate to support our operations;
- the potential benefits of strategic collaboration agreements and our ability to enter into strategic collaborations or other agreements with collaborators with development, regulatory and commercialization expertise;
- our financial performance;
- our tax positions and related tax liabilities;
- legal and regulatory developments in the United States and foreign countries;
- our ability to produce our products or product candidates with advantages in turnaround times or manufacturing cost; and
- developments relating to our competitors and our industry.

Forward-looking statements often contain words such as “will,” “may,” “should,” “could,” “expects,” “intends,” “plans,” “aims,” “anticipates,” “believes,” “estimates,” “predicts,” “potential,” “continue,” or the negative of these terms or other comparable terminology, although not all forward-looking statements contain these words. Although we believe that the expectations reflected in these forward-looking statements are reasonable, these statements relate to future events or our operational or financial performance, and involve risks, uncertainties, and other factors that may cause our actual results to differ materially from any future results expressed or implied by these forward-looking statements. Accordingly, you should not place undue reliance on these forward-looking statements. Factors that may cause actual results to differ materially from current expectations include, among other things, those listed under the section entitled “Risk Factors” and elsewhere in this Form 10-Q and under Part I, Item 1A. “Risk Factors” in our Annual Report on Form 10-K for the fiscal year ended December 31, 2025. If one or more of these risks or uncertainties occur, or if our underlying assumptions prove to be incorrect, actual results could differ materially from those expressed or implied by the forward-looking statements.

The forward-looking statements in this Form 10-Q represent our views as of the date of this Form 10-Q. We undertake no obligation to update any forward-looking statements, except as required by applicable securities law. You should therefore not rely on these forward-looking statements as representing our views as of any date subsequent to the date of this Form 10-Q. However, any further disclosures made on related subjects in our subsequent reports filed with the Securities and Exchange Commission should be consulted.

TRADEMARKS

This Form 10-Q contains references to our trademarks and to trademarks belonging to other entities. Solely for convenience, trademarks and trade names referred to may appear without the ® or ™ symbols, but such references are not intended to indicate that their respective owners will not assert, to the fullest extent under applicable law, their rights thereto. We do not intend our reference to other companies’ trade names or trademarks to imply a relationship with, or endorsement or sponsorship of us by, any other companies.

NOTE REGARDING COMPANY REFERENCES

Unless the context otherwise requires, the terms “Moderna,” the “Company,” “we,” “us” and “our” in this Form 10-Q refer to Moderna, Inc. and its consolidated subsidiaries.

ADDITIONAL INFORMATION

Our website, www.modernatx.com, including the Investor Relations section, www.investors.modernatx.com; and corporate blog www.modernatx.com/moderna-blog, and our Statements and Perspectives webpage, https://investors.modernatx.com/Statements--Perspectives/default.aspx; as well as our social media channels: Facebook, www.facebook.com/modernatx; X, www.x.com/moderna_tx (@moderna_tx); LinkedIn, www.linkedin.com/company/modernatx; Instagram (@moderna_tx); and Threads (@moderna_tx) contain a significant amount of information about us, including financial and other information for investors. We encourage investors to visit these websites and social media channels as information is frequently updated and new information is shared. Information contained on our website, corporate blog and social media channels shall not be deemed incorporated into, or be a part of, this Form 10-Q.

Table of Contents

		Page
PART I.		
Item 1.	Financial Statements (Unaudited)	5
	Condensed Consolidated Balance Sheets as of March 31, 2026 and December 31, 2025	5
	Condensed Consolidated Statements of Operations for the three months ended March 31, 2026 and 2025	6
	Condensed Consolidated Statements of Comprehensive Income (Loss) for the three months ended March 31, 2026 and 2025	7
	Condensed Consolidated Statements of Stockholders' Equity for the three months ended March 31, 2026 and 2025	8
	Condensed Consolidated Statements of Cash Flows for the three months ended March 31, 2026 and 2025	9
	Notes to Condensed Consolidated Financial Statements	10
Item 2.	Management’s Discussion and Analysis of Financial Condition and Results of Operations	28
Item 3.	Quantitative and Qualitative Disclosures about Market Risk	37
Item 4.	Controls and Procedures	37
PART II.		
Item 1.	Legal Proceedings	38
Item 1A.	Risk Factors	38
Item 2.	Unregistered Sales of Equity Securities and Use of Proceeds	39
Item 5.	Other Information	39
Item 6.	Exhibits	39
SIGNATURES		40

Item 1. Financial Statements

MODERNA, INC.
CONDENSED CONSOLIDATED BALANCE SHEETS
(Unaudited, in millions, except per share data)

	March 31, 2026	December 31, 2025
Assets		
Current assets:		
Cash and cash equivalents	$ 1,908	$ 2,595
Investments	3,297	3,204
Accounts receivable, net	71	184
Inventory	146	153
Prepaid expenses and other current assets	348	408
Total current assets	5,770	6,544
Investments, non-current	2,251	2,336
Property, plant and equipment, net	2,086	2,134
Right-of-use assets, operating leases	706	719
Other non-current assets	675	605
Total assets	$ 11,488	$ 12,338
Liabilities and Stockholders' Equity		
Current liabilities:		
Accounts payable	$ 161	$ 317
Accrued liabilities	1,912	1,386
Deferred revenue	102	99
Other current liabilities	220	185
Total current liabilities	2,395	1,987
Deferred revenue, non-current	154	153
Operating lease liabilities, non-current	645	653
Financing lease liabilities, non-current	13	20
Long-term debt	590	590
Other non-current liabilities	283	285
Total liabilities	4,080	3,688
Commitments and contingencies (Note 12)		
Stockholders' equity:		
Preferred stock, par value $0.0001; 162 shares authorized as of March 31, 2026 and December 31, 2025; no shares issued or outstanding at March 31, 2026 and December 31, 2025	—	—
Common stock, par value $0.0001; 1,600 shares authorized as of March 31, 2026 and December 31, 2025; 397 and 394 shares issued and outstanding as of March 31, 2026 and December 31, 2025, respectively	—	—
Additional paid-in capital	1,503	1,382
Accumulated other comprehensive income	25	45
Retained earnings	5,880	7,223
Total stockholders' equity	7,408	8,650
Total liabilities and stockholders' equity	$ 11,488	$ 12,338

The accompanying notes are an integral part of these unaudited condensed consolidated financial statements.

MODERNA, INC.
CONDENSED CONSOLIDATED STATEMENTS OF OPERATIONS
(Unaudited, in millions, except per share data)

	Three Months Ended March 31,	
	2026	**2025**
Revenue:		
Net product sales	$ 352	$ 86
Other revenue	37	22
Total revenue	389	108
Operating expenses:		
Cost of sales	955	90
Research and development	649	856
Selling, general and administrative	173	212
Total operating expenses	1,777	1,158
Loss from operations	(1,388)	(1,050)
Interest income	72	90
Other expense, net	(18)	(4)
Loss before income taxes	(1,334)	(964)
Provision for income taxes	9	7
Net loss	$ (1,343)	$ (971)
Net loss per share		
Basic and diluted	$ (3.40)	$ (2.52)
Weighted average common shares used in calculation of net loss per share		
Basic and diluted	395	386

The accompanying notes are an integral part of these unaudited condensed consolidated financial statements.

MODERNA, INC.
CONDENSED CONSOLIDATED STATEMENTS OF COMPREHENSIVE INCOME (LOSS)
(Unaudited, in millions)

	Three Months Ended March 31,	
	2026	2025
Net loss	$ (1,343)	$ (971)
Other comprehensive income, net of tax:		
Available-for-sale securities:		
Unrealized (losses) gains on available-for-sale securities	(17)	16
Less: net realized gains on available-for-sale securities reclassified in net loss	—	(1)
Net (decrease) increase from available-for-sale securities	(17)	15
Pension and postretirement obligation adjustments	—	2
(Losses) gains on foreign currency translation	(3)	3
Total other comprehensive (loss) income	(20)	20
Comprehensive loss	$ (1,363)	$ (951)

The accompanying notes are an integral part of these unaudited condensed consolidated financial statements.

MODERNA, INC.
CONDENSED CONSOLIDATED STATEMENTS OF STOCKHOLDERS' EQUITY
(Unaudited, in millions)

	Common Stock		Additional Paid-In Capital	Accumulated Other Comprehensive Income	Retained Earnings	Total Stockholders' Equity
	Shares	Amount				
Balance at December 31, 2025	394	$ —	$ 1,382	$ 45	$ 7,223	$ 8,650
Vesting of restricted common stock units	2	—	—	—	—	—
Exercise of options to purchase common stock	1	—	19	—	—	19
Tax payments related to net share settlements on equity awards	—	—	(2)	—	—	(2)
Stock-based compensation	—	—	104	—	—	104
Other comprehensive loss, net of tax	—	—	—	(20)	—	(20)
Net loss	—	—	—	—	(1,343)	(1,343)
Balance at March 31, 2026	397	$ —	$ 1,503	$ 25	$ 5,880	$ 7,408

	Common Stock		Additional Paid-In Capital	Accumulated Other Comprehensive (Loss) Income	Retained Earnings	Total Stockholders' Equity
	Shares	Amount				
Balance at December 31, 2024	386	$ —	$ 866	$ (10)	$ 10,045	$ 10,901
Vesting of restricted common stock	1	—	—	—	—	—
Exercise of options to purchase common stock	—	—	2	—	—	2
Tax payments related to net share settlements on equity awards	—	—	(1)	—	—	(1)
Stock-based compensation	—	—	115	—	—	115
Other comprehensive income, net of tax	—	—	—	20	—	20
Net loss	—	—	—	—	(971)	(971)
Balance at March 31, 2025	387	$ —	$ 982	$ 10	$ 9,074	$ 10,066

The accompanying notes are an integral part of these unaudited condensed consolidated financial statements.

MODERNA, INC.
CONDENSED CONSOLIDATED STATEMENTS OF CASH FLOWS
(Unaudited, in millions)

	Three Months Ended March 31,	
	2026	**2025**
Operating activities		
Net loss	$ (1,343)	$ (971)
Adjustments to reconcile net loss to net cash used in operating activities:		
Stock-based compensation	104	115
Depreciation and amortization	59	39
Amortization/accretion of investments	(11)	(19)
Loss on equity investments, net	2	8
Other non-cash items	6	2
Changes in assets and liabilities:		
Accounts receivable, net	114	280
Prepaid expenses and other assets	54	46
Inventory	6	(8)
Right-of-use assets, operating leases	11	9
Accounts payable	(120)	(156)
Accrued liabilities	464	(381)
Deferred revenue	5	(29)
Operating lease liabilities	(7)	(5)
Other liabilities	26	33
Net cash used in operating activities	(630)	(1,037)
Investing activities		
Purchases of marketable securities	(1,348)	(1,764)
Proceeds from maturities of marketable securities	732	1,933
Proceeds from sales of marketable securities	602	688
Purchases of property, plant and equipment	(62)	(117)
Purchase of intangible asset	—	(10)
Net cash (used in) provided by investing activities	(76)	730
Financing activities		
Proceeds from issuance of common stock through equity plans	19	3
Tax payments related to net share settlements on equity awards	(2)	(1)
Changes in financing lease liabilities	—	2
Net cash provided by financing activities	17	4
Effect of changes in exchange rates on cash and cash equivalents	1	—
Net decrease in cash, cash equivalents and restricted cash	(688)	(303)
Cash, cash equivalents and restricted cash, beginning of year	2,597	1,929
Cash, cash equivalents and restricted cash, end of period	$ 1,909	$ 1,626
Non-cash investing and financing activities		
Purchases of property and equipment included in accounts payable and accrued liabilities	$ 35	$ 50
Purchases of intangible asset included in accounts payable and accrued liabilities	74	—

The accompanying notes are an integral part of these unaudited condensed consolidated financial statements.

MODERNA, INC.
NOTES TO CONDENSED CONSOLIDATED FINANCIAL STATEMENTS
(Unaudited)

1. Description of the Business

Moderna, Inc. (collectively, with its consolidated subsidiaries, any of Moderna, we, us, our or the Company) is a biotechnology company advancing a new class of medicines made of messenger RNA (mRNA). mRNA medicines are designed to direct the body's cells to produce intracellular, membrane or secreted proteins that have a therapeutic or preventive benefit with the potential to address a broad spectrum of diseases. Our platform builds on continuous advances in basic and applied mRNA science, delivery technology and manufacturing, providing us the capability to pursue in parallel a robust pipeline of new development candidates. We are developing medicines across infectious disease vaccines, oncology therapeutics and rare disease therapeutics.

Since our founding in 2010, we have transformed from a research-stage company advancing programs in the field of mRNA to a commercial enterprise with a diverse clinical portfolio of vaccines and therapeutics across several modalities, a broad intellectual property portfolio and integrated manufacturing capabilities that allow for rapid clinical and commercial production at scale. As of March 31, 2026, we had three commercial products—Spikevax® and mNEXSPIKE®, our COVID vaccines, and mRESVIA®, our vaccine against respiratory syncytial virus (RSV). Additionally, we have a diverse development pipeline of 25 development candidates across our 35 development programs currently in clinical studies.

2. Summary of Basis of Presentation and Recent Accounting Standards

Basis of Presentation and Principles of Consolidation

The accompanying unaudited condensed consolidated financial statements that accompany these notes have been prepared in accordance with U.S. generally accepted accounting principles (GAAP) and applicable rules and regulations of the Securities and Exchange Commission (SEC) for interim financial reporting, consistent in all material respects with those applied in our Annual Report on Form 10-K for the year ended December 31, 2025 (2025 Form 10-K). Any reference in these notes to applicable guidance is meant to refer to the authoritative accounting principles generally accepted in the United States as found in the Accounting Standards Codification (ASC) and Accounting Standards Update (ASU) of the Financial Accounting Standards Board (FASB). This report should be read in conjunction with the audited consolidated financial statements in our 2025 Form 10-K.

The condensed consolidated financial statements include Moderna, Inc. and its subsidiaries. All intercompany transactions and balances have been eliminated in consolidation. The significant accounting policies used in the preparation of these condensed consolidated financial statements for the three months ended March 31, 2026 are consistent with those described in our 2025 Form 10-K. The results of operations for the three months ended March 31, 2026 are not necessarily indicative of the operating results to be expected for the full fiscal year or future operating periods. We anticipate seasonal fluctuations in demand for our COVID and RSV vaccines, with higher sales expected during the fall and winter seasons.

Use of Estimates

We have made estimates and judgments affecting the amounts reported in our condensed consolidated financial statements and the accompanying notes. We base our estimates on historical experience and various relevant assumptions that we believe to be reasonable under the circumstances, the results of which form the basis for making judgments about the carrying values of assets and liabilities at the date of the financial statements and the reported amounts of revenues and expenses during the reporting periods that are not readily apparent from other sources. Changes in our estimates are recorded in the financial results of the period in which the new information becomes available. The actual results that we experience may differ materially from our estimates.

Comprehensive Income (Loss)

Comprehensive income (loss) includes net income (loss) and other comprehensive income/loss for the period. Other comprehensive income/loss consists of unrealized gains/losses on our investments, derivatives designated as hedging instruments, foreign currency translation, and pension and postretirement obligation adjustments. Total comprehensive income (loss) for all periods presented has been disclosed in the condensed consolidated statements of comprehensive income (loss).

The components of accumulated other comprehensive income (loss) for the three months ended March 31, 2026 were as follows (in millions):

	Unrealized Gains (losses) on Available-for-Sale Debt Securities	Pension and Postretirement Obligation Adjustments	Gains (Losses) on Foreign Currency Translation	Total
Accumulated other comprehensive income, balance at December 31, 2025	$ 38	$ 4	$ 3	$ 45
Other comprehensive loss	(17)	—	(3)	(20)
Accumulated other comprehensive income, balance at March 31, 2026	$ 21	$ 4	$ —	$ 25

Restricted Cash

We include our restricted cash balance in the cash, cash equivalents and restricted cash reconciliation of operating, investing and financing activities in the condensed consolidated statements of cash flows.

The following table provides a reconciliation of cash, cash equivalents and restricted cash in the condensed consolidated balance sheets that sum to the total of the same such amounts shown in the condensed consolidated statements of cash flows (in millions):

	March 31,	
	2026	2025
Cash and cash equivalents	$ 1,908	$ 1,623
Restricted cash[(1)]	—	1
Restricted cash, non-current[(2)]	1	2
Total cash, cash equivalents and restricted cash shown in the condensed consolidated statements of cash flows	$ 1,909	$ 1,626

[(1)]Included in prepaid expenses and other current assets in the condensed consolidated balance sheets.

[(2)]Included in other non-current assets in the condensed consolidated balance sheets.

Recently Issued Accounting Standards

From time to time, new accounting pronouncements are issued by the FASB or other standard setting bodies and adopted by us as of the specified effective date. Except as noted below, we believe that the impact of recently issued standards that are not yet effective will not have a material impact on our condensed consolidated financial statements and disclosures.

In November 2024, the FASB issued ASU 2024-03, Income Statement - Reporting Comprehensive Income - Expense Disaggregation Disclosures (Subtopic 220-40): Disaggregation of Income Statement Expenses. This ASU requires entities to disclose, on an annual and interim basis, disaggregated information in the footnotes related to certain expense categories included in income statement line items. Specifically, entities are expected to provide tabular disclosures for prescribed categories such as inventory purchases, employee compensation, depreciation, and intangible asset amortization for each relevant expense caption. The standard also requires disclosure of total selling expenses and a definition of those expenses in annual filings. Any remaining amounts not quantitatively disclosed are expected to be described qualitatively. This ASU is effective for fiscal years beginning after December 15, 2026 and interim periods beginning after December 15, 2027. Early adoption permitted, and the standard may be applied on a prospective or retrospective basis. We are currently assessing the impact that this new accounting standard will have on our consolidated financial statement disclosures.

In September 2025, the FASB issued ASU No. 2025-06, Intangibles - Goodwill and Other - Internal-Use Software (Subtopic 350-40): Targeted Improvements to the Accounting for Internal-Use Software. This ASU makes targeted amendments to the accounting for and disclosure of software costs under ASC 350-40. The amendments modernize the guidance to reflect current software development practices, including nonlinear development approaches, and remove references to "development stages." Under the ASU, the following two criteria must be met for entities to begin capitalizing software costs: (1) management, with the relevant authority, implicitly or explicitly authorizes and commits to funding a computer software project, and (2) it is probable that the project will be completed and the software will be used to perform the function intended (referred to as the "probable-to-complete recognition threshold"). The ASU clarifies that this threshold would not be met when there is "significant uncertainty associated with the development activities of the software (referred to as 'significant development uncertainty')." The new standard is effective for all entities for annual reporting periods beginning after December 15, 2027, and for interim reporting periods within those fiscal years. Early adoption is permitted, and entities may apply the amendments prospectively, retrospectively, or using a modified prospective transition approach. We are currently evaluating the impact that this new accounting standard will have on our consolidated financial statements and disclosures.

In September 2025, the FASB issued ASU No. 2025-07, Derivatives and Hedging (Topic 815) and Revenue from Contracts with Customers (Topic 606): Derivatives Scope Refinements and Scope Clarification for Share-Based Noncash Consideration From a Customer in a Revenue Contract. This ASU expands the scope exceptions in the derivatives guidance to exclude certain non-exchange-traded contracts with underlyings based on the operations or activities of one of the parties to the contract, including the occurrence or nonoccurrence of an event specific to those operations or activities. The ASU also clarifies that share-based noncash consideration received from a customer in exchange for goods or services should be accounted for as noncash consideration under ASC 606 unless and until the entity's right to receive or retain such consideration becomes unconditional. The ASU is effective for annual reporting periods beginning after December 15, 2026, including interim periods within those annual reporting periods. Early adoption is permitted. We are currently evaluating the impact that adoption of this new accounting standard will have on our consolidated financial statements and disclosures.

In December 2025, the FASB issued ASU No. 2025-10, Government Grants (Topic 832): Accounting for Government Grants Received by Business Entities. This ASU establishes guidance on the recognition, measurement, presentation, and disclosure of government grants received by business entities. The guidance defines a government grant as a transfer of a monetary asset or a tangible nonmonetary asset from a government to a business entity other than in an exchange transaction and excludes transactions within the scope of other U.S. GAAP. Under the ASU, government grants are classified as either grants related to an asset or grants related to income, and recognition is permitted only when it is probable that the entity will comply with the conditions attached to the grant and that the grant will be received. The ASU permits alternative presentation approaches depending on the nature of the grant and requires disclosures regarding the nature of the grant, affected financial statement line items, and significant terms and conditions. The ASU is effective for public business entities for annual reporting periods beginning after December 15, 2028, including interim periods within those annual reporting periods. Early adoption is permitted, and the standard may be applied on a modified prospective, modified retrospective, or full retrospective basis. We are currently evaluating the impact of this new accounting standard and do not expect its adoption to have a material impact on our consolidated financial statements and disclosures.

3. Net Product Sales

Net product sales by customer geographic location were as follows (in millions):

	Three Months Ended March 31,	
	2026	2025
United States	$ 73	$ 31
Europe	239	—
Rest of world	40	55
Total	$ 352	$ 86

Net product sales by product type were as follows (in millions):

	Three Months Ended March 31,	
	2026	2025
COVID[(1)]	$ 345	$ 84
RSV	7	2
Total	$ 352	$ 86

[(1)]Includes sales of Spikevax and mNEXSPIKE.

As of March 31, 2026, we have three commercial products, our COVID vaccines, Spikevax and mNEXSPIKE, and our RSV vaccine, mRESVIA. mRESVIA was approved by the FDA in May 2024 for adults aged 60 years and older, and in June 2025, the approved use was expanded to include adults aged 18 through 59 years who are at increased risk for lower respiratory tract disease caused by RSV. In May 2025, mNEXSPIKE was approved for use in adults aged 65 years and older, as well as individuals aged 12 through 64 years with at least one underlying risk factor. We launched commercial sales of mNEXSPIKE in the third quarter of 2025.

We sell our COVID and RSV vaccines to the commercial market as well as to foreign governments and international organizations. In the U.S., our COVID and RSV vaccines are sold primarily to wholesalers and distributors, and to a lesser extent, directly to retailers and healthcare providers. Wholesalers and distributors typically do not make upfront payments to us. Net product sales are recognized net of estimated wholesaler chargebacks, invoice discounts for prompt payments and pre-orders, provisions for sales returns and government rebates, and other related deductions.

The following table summarizes product sales provision adjustments for the periods presented (in millions):

	Three Months Ended March 31,	
	2026	2025
Gross product sales	$ 350	$ 105
Product sales provision:		
Wholesaler chargebacks, discounts and fees	11	(22)
Returns, rebates and other fees	(9)	3
Total product sales provision adjustments	$ 2	$ (19)
Net product sales	$ 352	$ 86

The following table summarizes the activities related to product sales provision recorded as accrued liabilities for the three months ended March 31, 2026 (in millions):

	Returns and other fees
Balance at December 31, 2025	$ (509)
Provision related to sales made in current period	(9)
Provision related to sales made in prior periods	—
Payments and returns related to sales made in current period	—
Payments and returns related to sales made in prior year	48
Balance at March 31, 2026	$ (470)

4. Other Revenue

The following table summarizes other revenue for the periods presented (in millions):

	Three Months Ended March 31,	
	2026	2025
Stand-ready manufacturing revenue	$ 32	$ 12
Collaboration revenue (Note 5)	5	1
Grant revenue	—	1
Licensing and royalty revenue	—	8
Total other revenue	$ 37	$ 22

5. Collaboration Agreements and Research and Development Funding Arrangement

Merck – Individualized Neoantigen Therapy (Intismeran Autogene)

In June 2016, we entered into a Collaboration and License Agreement, which was subsequently amended in 2018, with Merck & Co., Inc. (Merck) for the development and commercialization of individualized neoantigen therapy (INT), which has been assigned the generic name intismeran autogene.

In September 2022, Merck exercised its option for intismeran, including mRNA-4157, pursuant to the terms of the agreement and in October 2022 paid us an option exercise fee of $250 million. Following this exercise, the Merck Participation Term commenced. Pursuant to the agreement, we and Merck have agreed to collaborate on development and potential commercialization of intismeran, with costs and any profits or losses generally shared equally on a worldwide basis, subject to certain exceptions as outlined in the agreement. During the development phase, we are primarily responsible for process development and the manufacture of intismeran materials, while Merck generally leads clinical trials. We concluded that the collaboration arrangement under the Merck Participation Term is within the scope of ASC 808. For the three months ended March 31, 2026 and 2025, we recognized expenses, net of Merck's reimbursements, of $101 million and $104 million, respectively, related to the INT collaboration under the Merck Participation Term. Additionally, for the three months ended March 31, 2026 and 2025, the net cost recovery for capital expenditures was $1 million and $12 million, respectively. These amounts were applied to reduce the capitalized cost of the assets.

Recordati –Propionic Acidemia Therapeutic

In January 2026, we entered into a collaboration agreement with Recordati S.p.A. (Recordati) to advance the development and commercialization of our investigational propionic acidemia therapeutic, mRNA-3927. Under the terms of the agreement, we granted Recordati an exclusive, royalty-bearing license to develop and commercialize the licensed product (mRNA-3927) worldwide, we will perform and continue to lead clinical development activities through approval, and Recordati will lead global commercialization upon approval. The transaction closed on March 16, 2026, and we are entitled to receive a $50 million upfront payment. As of March 31, 2026, we recorded a receivable and deferred revenue for this amount, which is expected to be recognized as collaboration revenue over time. In addition, we are eligible to receive up to $110 million in development and regulatory milestones, as well as additional commercial and sales milestones and tiered royalties on net sales.

We have other collaborative and licensing arrangements that we do not consider to be individually significant to our business at this time. Pursuant to these agreements, we may be required to make upfront payments and payments upon achievement of various development, regulatory and commercial milestones, which in the aggregate could be significant. Future milestone payments, if any, will be reflected in our consolidated financial statements when the corresponding events have occurred. In addition, we may be required to pay significant royalties on future sales if products related to these arrangements are commercialized.

Development and Commercialization Funding Arrangement with Blackstone Life Sciences (Blackstone)

In March 2024, we entered into a development and commercialization funding arrangement with Blackstone, under which Blackstone has committed to providing up to $750 million in funding to us. This funding supports the development of our investigational mRNA-based influenza vaccine. Contingent upon regulatory approval in the U.S. and only if the approval is dependent on data from the funded activities, Blackstone will be entitled to receive low single-digit percentage royalties and up to $750 million in sales milestone payments. These payments are based on net sales of our future influenza and combination vaccines, with sales milestone payments contingent upon achieving specified cumulative net sales targets.

Given the substantive transfer of financial risk to Blackstone, we account for this arrangement as an obligation to conduct research and development activities. The funding is recognized as a reduction to the expenses of our mRNA-based influenza program. This reduction is recognized proportionally as the related costs are incurred, based on an input method. For the three months ended March 31, 2026 and 2025, we recorded expense reductions of $50 million and $90 million, respectively.

6. Financial Instruments

Cash and Cash Equivalents and Investments

The following tables summarize our cash, cash equivalents, and available-for-sale securities by significant investment category as of March 31, 2026 and December 31, 2025 (in millions):

	March 31, 2026						
	Amortized Cost	Unrealized Gains	Unrealized Losses	Estimated Fair Value	Cash and Cash Equivalents	Current Marketable Securities	Non-Current Marketable Securities
Cash and cash equivalents	$ 1,908	$ —	$ —	$ 1,908	$ 1,908	$ —	$ —
Available-for-sale:							
Certificates of deposit	61	—	—	61	—	61	—
U.S. treasury bills	610	—	—	610	—	610	—
U.S. treasury notes	2,230	1	(5)	2,226	—	1,233	993
Corporate debt securities	2,599	2	(6)	2,595	—	1,391	1,204
Government debt securities	56	—	—	56	—	2	54
Total	$ 7,464	$ 3	$ (11)	$ 7,456	$ 1,908	$ 3,297	$ 2,251

	December 31, 2025						
	Amortized Cost	Unrealized Gains	Unrealized Losses	Estimated Fair Value	Cash and Cash Equivalents	Current Marketable Securities	Non-Current Marketable Securities
Cash and cash equivalents	$ 2,595	$ —	$ —	$ 2,595	$ 2,595	$ —	$ —
Available-for-sale:							
Certificates of deposit	91	—	—	91	—	86	5
U.S. treasury bills	653	—	—	653	—	653	—
U.S. treasury notes	2,188	5	(3)	2,190	—	1,185	1,005
Corporate debt securities	2,535	6	(1)	2,540	—	1,250	1,290
Government debt securities	66	—	—	66	—	30	36
Total	$ 8,128	$ 11	$ (4)	$ 8,135	$ 2,595	$ 3,204	$ 2,336

The amortized cost and estimated fair value of available-for-sale securities by contractual maturity as of March 31, 2026 and December 31, 2025 were as follows (in millions):

	March 31, 2026	
	Amortized Cost	Estimated Fair Value
Due in one year or less	$ 3,297	$ 3,297
Due after one year through five years	2,259	2,251
Total	$ 5,556	$ 5,548

	December 31, 2025	
	Amortized Cost	Estimated Fair Value
Due in one year or less	$ 3,199	$ 3,204
Due after one year through five years	2,334	2,336
Total	$ 5,533	$ 5,540

In accordance with our investment policy, we place investments in investment grade securities with high credit quality issuers, and generally limit the amount of credit exposure to any one issuer. We evaluate securities for impairment at the end of each reporting period. Impairment is evaluated considering numerous factors, and their relative significance varies depending on the situation.

Factors considered include whether a decline in fair value below the amortized cost basis is due to credit-related factors or non-credit-related factors, the financial condition and near-term prospects of the issuer, and our intent and ability to hold the investment to allow for an anticipated recovery in fair value. Any impairment that is not credit related is recognized in other comprehensive loss, net of applicable taxes. A credit-related impairment is recognized as an allowance on the balance sheet with a corresponding adjustment to earnings. We did not recognize any impairment charges related to available-for-sale securities for the three months ended March 31, 2026 and 2025. We did not record any credit-related allowance for available-for-sale securities as of March 31, 2026 and December 31, 2025.

The following table summarizes the amount of gross unrealized losses and the estimated fair value for our available-for-sale securities in an unrealized loss position by the length of time the securities have been in an unrealized loss position as of March 31, 2026 and December 31, 2025 (in millions):

	Less than 12 Months		12 Months or More		Total	
	Gross Unrealized Losses	Estimated Fair Value	Gross Unrealized Losses	Estimated Fair Value	Gross Unrealized Losses	Estimated Fair Value
As of March 31, 2026:						
U.S. treasury bills	$ —	$ 1,023	$ —	$ —	$ —	$ 1,023
U.S. treasury notes	(3)	960	(2)	236	(5)	1,196
Corporate debt securities	(6)	1,562	—	35	(6)	1,597
Government debt securities	—	56	—	—	—	56
Total	$ (9)	$ 3,601	$ (2)	$ 271	$ (11)	$ 3,872
As of December 31, 2025:						
U.S. treasury bills	$ —	$ 111	$ —	$ —	$ —	$ 111
U.S. treasury notes	(1)	176	(2)	235	(3)	411
Corporate debt securities	—	608	(1)	40	(1)	648
Government debt securities	—	36	—	8	—	44
Total	$ (1)	$ 931	$ (3)	$ 283	$ (4)	$ 1,214

As of March 31, 2026 and December 31, 2025, we held 292 and 108 available-for-sale securities, respectively, out of our total investment portfolio that were in a continuous unrealized loss position. We neither intend to sell these investments, nor do we believe that we are more-likely-than-not to conclude we will have to sell them before recovery of their carrying values. We also believe that we will be able to collect both principal and interest amounts due to us at maturity.

Assets and Liabilities Measured at Fair Value on a Recurring Basis

The following fair value hierarchy is used to classify assets and liabilities based on the observable inputs and unobservable inputs used to value the assets and liabilities:

- Level 1: Unadjusted quoted prices in active markets that are accessible at the measurement date for identical, unrestricted assets or liabilities;
- Level 2: Quoted prices for similar assets and liabilities in active markets, quoted prices in markets that are not active, or inputs which are observable, either directly or indirectly, for substantially the full term of the asset or liability; or
- Level 3: Prices or valuation techniques that require inputs that are both significant to the fair value measurement and unobservable (i.e., supported by little or no market activity).

The following tables summarize our financial assets and liabilities measured at fair value on a recurring basis as of March 31, 2026 and December 31, 2025 (in millions):

	Fair value at March 31, 2026	Fair Value Measurement Using	
		Level 1	Level 2
Assets:			
Money market funds	$ 595	$ 595	$ —
Certificates of deposit	61	—	61
U.S. treasury bills	1,141	—	1,141
U.S. treasury notes	2,226	—	2,226
Corporate debt securities	2,978	—	2,978
Government debt securities	56	—	56
Equity investments[1]	2	2	—
Derivative instruments	8	—	8
Total	$ 7,067	$ 597	$ 6,470

	Fair value at December 31, 2025	Fair Value Measurement Using	
		Level 1	Level 2
Assets:			
Money market funds	$ 963	$ 963	$ —
Certificates of deposit	91	—	91
U.S. treasury bills	1,445	—	1,445
U.S. treasury notes	2,190	—	2,190
Corporate debt securities	3,163	—	3,163
Government debt securities	66	—	66
Equity Investments[1]	6	6	—
Derivative instruments	1	—	1
Total	$ 7,925	$ 969	$ 6,956
Liabilities:			
Derivative instruments	$ 4	$ —	$ 4

[1]Investments in publicly traded equity securities with readily determinable fair values are recorded at quoted market prices for identical securities, with changes in fair value recorded in other expense, net, in our condensed consolidated statements of operations.

As of March 31, 2026 and December 31, 2025, we did not have non-financial assets or liabilities measured at fair value on a recurring basis and did not have any Level 3 financial assets or financial liabilities.

For the three months ended March 31, 2026 and 2025, we recognized net losses of $2 million and $8 million, respectively, on equity investments from changes in fair value of the securities.

Fair Value of Other Financial Instruments

We estimate the fair value of our term loan using Level 2 inputs. The fair value of the term loan approximates its carrying value as of March 31, 2026 and December 31, 2025, as the instrument bears interest at a variable rate that reflects current market rates. See Note 11 for additional information.

7. Inventory

Inventory as of March 31, 2026 and December 31, 2025 consisted of the following (in millions):

	March 31, 2026	December 31, 2025
Raw materials	$ 94	$ 91
Work in progress	38	29
Finished goods	14	33
Total inventory	$ 146	$ 153
Inventory, non-current[1]	$ 117	$ 114

[1] Consisted of raw materials with an anticipated consumption beyond one year. Inventory, non-current is included in other non-current assets in the condensed consolidated balance sheets.

Inventory write-downs as a result of excess, obsolescence, scrap or other reasons, and losses on firm purchase commitments are recorded as a component of cost of sales in our condensed consolidated statements of operations. For the three months ended March 31, 2026 and 2025, inventory write-downs were $38 million and $42 million, respectively. For the three months ended March 31, 2026, there were no losses on firm purchase commitments. For the three months ended March 31, 2025, losses on firm purchase commitments was $10 million.

Inventory write-downs were mainly related to inventory in excess of expected demand, shelf-life expiration and other inventory adjustments. Losses on firm purchase commitments were primarily related to excess raw material purchase commitments that will expire before the anticipated consumption of those raw materials.

As of March 31, 2026 and December 31, 2025, we had inventory on hand of $263 million and $267 million, respectively, inclusive of inventory for our COVID and RSV vaccines. Our raw materials and work-in-progress inventory have variable shelf lives. We expect that the majority of this inventory will be consumed over the next three years. The shelf life of Spikevax is nine to twelve months. mNEXSPIKE has a shelf life of twelve months. The shelf life of mRESVIA, our RSV vaccine, is eighteen months.

8. Property, Plant and Equipment, Net

Property, plant and equipment, net, as of March 31, 2026 and December 31, 2025 consisted of the following (in millions):

	March 31, 2026	December 31, 2025
Land and land improvements	$ 77	$ 78
Building and building improvements	1,216	1,183
Manufacturing and laboratory equipment	543	542
Leasehold improvements	408	403
Furniture, fixtures and other	40	39
Computer equipment and software	196	196
Construction in progress	268	298
Right-of-use assets, financing (Note 10)	132	132
Total	2,880	2,871
Less: Accumulated depreciation	(794)	(737)
Property, plant and equipment, net	$ 2,086	$ 2,134

Depreciation and amortization expense related to property, plant and equipment for the three months ended March 31, 2026 and 2025 was $56 million and $38 million, respectively.

9. Other Balance Sheet Components

Accounts Receivable, net

Accounts receivable, net, as of March 31, 2026 and December 31, 2025 consisted of the following (in millions):

	March 31, 2026	December 31, 2025
Accounts receivable	$ 203	$ 368
Less: Wholesalers chargebacks, discounts and fees	(126)	(181)
Less: Allowance for doubtful accounts	(6)	(3)
Accounts receivable, net	$ 71	$ 184

Prepaid Expenses and Other Current Assets

Prepaid expenses and other current assets, as of March 31, 2026 and December 31, 2025 consisted of the following (in millions):

	March 31, 2026	December 31, 2025
Prepaid services	$ 131	$ 169
Down payments and prepayments related to manufacturing and materials	55	61
Interest receivable	39	42
Prepaid income tax and income tax receivable	35	41
Value added tax receivable	29	37
Research and development funding receivable (Note 5)	7	—
Collaboration receivable	1	13
Other current assets	51	45
Prepaid expenses and other current assets	$ 348	$ 408

Other Non-Current Assets

Other non-current assets, as of March 31, 2026 and December 31, 2025 consisted of the following (in millions):

	March 31, 2026	December 31, 2025
Income tax receivable, non-current	$ 161	$ 161
Inventory, non-current[(1)]	117	114
Finite-lived intangible assets, net[(2)]	116	45
Down payments and prepayments, non-current	97	100
Deferred tax assets	82	81
Goodwill	52	52
Other	50	52
Other non-current assets	$ 675	$ 605

[(1)] Consisted of raw materials with an anticipated consumption beyond one year.
[(2)] Includes a $72 million intangible asset related to a license recognized in connection with the settlement agreement with Arbutus and Genevant. See Note 12 for additional details.

Accrued Liabilities

Accrued liabilities, as of March 31, 2026 and December 31, 2025 consisted of the following (in millions):

	March 31, 2026	December 31, 2025
Royalties[1]	$ 967	$ 31
Provisions related to product sales (Note 3)	470	509
Compensation-related	108	420
Other external goods and services	82	57
Development operations	81	106
Manufacturing	79	140
Clinical trials	46	20
Property, plant and equipment	31	45
Raw materials	27	30
Commercial	16	23
Loss on future firm purchase commitments[2]	5	5
Accrued liabilities	$ 1,912	$ 1,386

[1] Includes $950 million related to the litigation settlement with Arbutus and Genevant (Note 12).
[2] Related to losses that are expected to arise from firm, non-cancellable, commitments for future raw material purchases (Note 7).

Other Current Liabilities

Other current liabilities, as of March 31, 2026 and December 31, 2025 consisted of the following (in millions):

	March 31, 2026	December 31, 2025
Estimated reimbursements to wholesalers and distributors	$ 111	$ 84
Lease liabilities - financing (Note 10)	31	25
Lease liabilities - operating (Note 10)	19	17
Research and development funding liability (Note 5)	—	43
Other	59	16
Other current liabilities	$ 220	$ 185

Other Non-Current Liabilities

Other non-current liabilities, as of March 31, 2026 and December 31, 2025 consisted of the following (in millions):

	March 31, 2026	December 31, 2025
Tax liabilities	$ 253	$ 249
Other	30	36
Other non-current liabilities	$ 283	$ 285

Deferred Revenue

The following table summarizes the activities in deferred revenue for the three months ended March 31, 2026 (in millions):

	December 31, 2025	Additions	Deductions	March 31, 2026
Other revenue	$ 145	$ 53	$ (37)	$ 161
Net product sales	107	—	(12)	95
Total deferred revenue	$ 252	$ 53	$ (49)	$ 256

10. Leases

We have entered into various long-term, non-cancelable lease arrangements for our facilities and equipment, expiring at various times through 2039. Certain of these arrangements have free rent periods or escalating rent payment provisions. We recognize lease costs under such arrangements on a straight-line basis over the life of the lease. We lease various parcels of land, office, lab, and manufacturing spaces across the globe for our business operations.

Our primary leased campus is our Moderna Science Center (MSC), located in Cambridge, which serves as our headquarters. The MSC, comprising approximately 462,000 square feet, includes our principal executive office and additional office and laboratory spaces. The MSC lease commenced in the third quarter of 2023 and has a term of 15 years, with options for two additional seven-year extensions.

Operating and financing lease right-of-use assets and lease liabilities as of March 31, 2026 and December 31, 2025 were as follows (in millions):

	March 31, 2026	December 31, 2025
Assets:		
Right-of-use assets, operating, net[1] [2]	$ 706	$ 719
Right-of-use assets, financing, net[3] [4]	36	42
Total	$ 742	$ 761
Liabilities:		
Current:		
Operating lease liabilities[5]	$ 19	$ 17
Financing lease liabilities[5]	31	25
Total current lease liabilities	50	42
Non-current:		
Operating lease liabilities, non-current	645	653
Financing lease liabilities, non-current	13	20
Total non-current lease liabilities	658	673
Total	$ 708	$ 715

[1] These assets are real estate related assets, which include land, office, manufacturing, and laboratory spaces.
[2] Net of accumulated amortization.
[3] These assets are related to contract manufacturing service agreements.
[4] Included in property, plant and equipment in the condensed consolidated balance sheets, net of accumulated depreciation.
[5] Included in other current liabilities in the condensed consolidated balance sheets.

Future minimum lease payments under our non-cancelable lease agreements as of March 31, 2026, were as follows (in millions):

Fiscal Year	Operating Leases	Financing Leases
2026 (remainder of the year)	$ 47	$ 26
2027	78	20
2028	81	—
2029	82	—
2030	80	—
Thereafter	679	—
Total minimum lease payments	1,047	46
Less amounts representing interest or imputed interest	(383)	(2)
Present value of lease liabilities	$ 664	$ 44

11. Credit Agreement

In November 2025, we entered into a Credit and Guaranty Agreement (the Credit Agreement) with lenders led by Ares Capital Corporation, as administrative agent. The Credit Agreement provides for a senior secured term loan facility with aggregate term loan commitments of $1.5 billion, consisting of a $600 million initial term loan, which was funded at closing, and $900 million of delayed draw term loan commitments. The initial term loan matures on November 24, 2030. The delayed draw term loan commitments consist of (1) a $400 million delayed draw term loan facility (DDTL-1), which is available, subject to customary conditions, through November 24, 2027, and (2) a $500 million delayed draw term loan facility (DDTL-2), which is available, subject to customary conditions and the achievement of specified regulatory approval milestones for certain product candidates, through November 24, 2028.

Borrowings under the Credit Agreement bear interest at a variable rate equal to, at our option, (i) Term SOFR plus a margin of 5.50% or (ii) a base rate plus a margin of 4.50%. The base rate is calculated as the highest of (a) the Wall Street Journal prime rate, (b) the federal funds rate plus one half of one percent and (c) Term SOFR plus one percent. We are also required to pay commitment fees on the undrawn portions of DDTL-1 and DDTL-2. The interest rate applicable to the initial term loan was approximately 9.17% and 9.38% as of March 31, 2026 and December 31, 2025, respectively.

The obligations under the Credit Agreement are guaranteed by certain of our subsidiaries and are secured by a first-priority lien on substantially all of our assets, in each case subject to customary exceptions and limitations. The Credit Agreement is subject to compliance with customary representations and warranties, affirmative covenants, restrictive covenants and events of default. The restrictive covenants, subject to specified limitations and exceptions, limit, among other things, our ability to incur additional indebtedness and liens, make certain investments, engage in certain fundamental changes, dispose of assets and make restricted payments. Events of default under the Credit Agreement include, among others, nonpayment of principal, interest or other amounts when due, failure to comply with covenants (subject to applicable notice and cure periods), breaches of certain representations and warranties, the occurrence of certain significant adverse events and certain insolvency-related events. The Credit Agreement also includes a financial covenant requiring us to maintain minimum cash and cash equivalents (as defined in the Credit Agreement, which primarily consist of our cash, cash equivalents, and available-for-sale securities) as of the last business day of each week of at least $500 million, increasing to $750 million if more than $1.0 billion is drawn under the Credit Agreement. The financial covenant is not required to be tested at any time that the trailing 30-day average market capitalization of the Company exceeds $5.0 billion and is subject to a customary equity cure. As of March 31, 2026 and December 31, 2025, we were in compliance with the applicable terms and covenants under the Credit Agreement.

As of March 31, 2026 and December 31, 2025, the initial term loan had an outstanding principal balance of $600 million and a carrying amount of $590 million at each date, net of unamortized original issue discount and debt issuance costs, which was classified as long-term debt in our condensed consolidated balance sheet. No amounts had been drawn under DDTL-1 or DDTL-2. The principal amount of $600 million is due in full at maturity, and no principal payments are required prior to that date.

12. Commitments and Contingencies

Legal Proceedings

We are a party to various legal proceedings and claims. Accruals are recognized for legal matters when a loss is both probable and reasonably estimable. As of March 31, 2026, no material contingent liabilities have been recognized. If a material loss is reasonably possible and we can estimate the amount or range of the loss, we disclose such information. Unless otherwise noted, either the outcome of these matters is not expected to be material, or the potential loss cannot be reasonably estimated.

From time to time, we may be a party to litigation, arbitration, or other legal proceedings in the course of our business. The outcome of such matters is inherently uncertain and often involves significant judgment in assessing risk and estimating potential exposure. While we do not currently expect any pending proceedings to have a material adverse effect on our financial position, results of operations, or cash flows, there can be no assurance that future developments will not have a material impact.

On March 3, 2026, we entered into a settlement agreement with Arbutus Biopharma Corporation (Arbutus) and Genevant Sciences GmbH (Genevant, and with Arbutus, Arbutus/Genevant) resolving all litigation worldwide, including between the parties in the U.S. District Court for the District of Delaware. The settlement resolves all worldwide Arbutus/Genevant litigation related to Spikevax and mRESVIA and provides certainty going forward for our full infectious disease portfolio, including mNEXSPIKE, mCOMBRIAX and our future vaccine pipeline, with no future royalties owed. Under the terms of the agreement, we agreed to make a lump sum payment of $950 million, which is payable in the third quarter of 2026. In the three months ended March 31, 2026, we recorded $876 million in cost of sales and capitalized $74 million as an intangible asset on the settlement date, representing the estimated value attributed to a license to certain intellectual property rights, which is being amortized on a straight-line basis over an estimated useful life of approximately three years. As of March 31, 2026, the carrying value of the intangible asset was $72 million. Consistent with the terms of the settlement agreement, we have appealed the District Court's decision related to 28 U.S.C. § 1498 to the Federal Circuit Court of Appeals and could be required to make an additional payment of up to $1.3 billion pending on the outcome. As of March 31, 2026, no accrual has been recorded for this amount as a loss is not considered probable.

The following summarizes our other significant legal proceedings and matters outstanding as of March 31, 2026.

We have brought patent-infringement actions against Pfizer Inc. (Pfizer), BioNTech SE (BioNTech) and related entities in the U.S., Germany, the Netherlands, the UK, Ireland and Belgium concerning our mRNA platform technology and disease-specific vaccine designs. Pfizer and BioNTech have commenced actions or asserted defenses seeking to revoke our patents in these jurisdictions.

GlaxoSmithKline Biologicals SA (GSK) has filed two complaints against us in the U.S. District Court for the District of Delaware asserting certain patents owned by GSK. GSK has also filed two patent-infringement lawsuits against us in the UPC concerning liposomes and modified liposomes for RNA delivery.

Northwestern University has filed a complaint against us in the U.S. District Court for the District of Delaware asserting U.S. patents concerning lipid nanoparticle technology.

Bayer CropSciences LLC, Monsanto Company, and Monsanto Technology, LLC have filed a complaint against us in the U.S. District Court for the District of Delaware asserting a U.S. patent directed to methods of modifying gene sequences.

mNG Bio, LLC has filed a complaint against us in the U.S District Court for the District of Massachusetts asserting a U.S. patent directed to a yellow-green fluorescent protein.

BioNTech SE has filed a complaint against us in the U.S. District Court for the District of Delaware asserting a U.S. patent directed to modified mRNA compositions encoding a spike protein fragment.

CureVac SE and CureVac Manufacturing GmbH have filed a complaint against us in the U.S. District Court for the District of Delaware asserting U.S. patents directed to RNA production methods and compositions with RNA encoding modified SARS-CoV-2 spike protein.

We are subject to shareholder class action and shareholder derivative litigation pending in the U.S. District Court for the District of Massachusetts related to statements about our RSV vaccine (mRNA-1345).

Indemnification Obligations

As permitted under Delaware law, we indemnify our officers, directors, and employees for certain events, occurrences while the officer, or director is, or was, serving at our request in such capacity. The term of the indemnification is for the officer's or director's lifetime.

We have standard indemnification arrangements in our leases for laboratory and office space that require us to indemnify the landlord against any liability for injury, loss, accident, or damage from any claims, actions, proceedings, or costs resulting from certain acts, breaches, violations, or non-performance under our leases.

We enter into indemnification provisions under our agreements with counterparties in the ordinary course of business, typically with business partners, contractors, clinical sites and customers. Under these provisions, we generally indemnify and hold harmless the indemnified party for losses suffered or incurred by the indemnified party as a result of our activities. These indemnification provisions generally survive termination of the underlying agreement. The maximum potential amount of future payments we could be required to make under these indemnification provisions is unlimited.

Through the three months ended March 31, 2026 and the year ended December 31, 2025, we had not experienced any material losses related to these indemnification obligations, and no material claims were outstanding. We do not expect significant claims related to these indemnification obligations and, consequently, concluded that the fair value of these obligations is negligible, and no related reserves were established.

Licenses to Patented Technology

We have patent license agreements with Cellscript, LLC and its affiliate, mRNA RiboTherapeutics, Inc., and the National Institute of Allergy and Infectious Diseases (NIAID), an Institute of the National Institutes of Health (NIH). Under these agreements, we are required to pay royalties and certain milestone payments. For further information on our licensing and royalty payments, please refer to our 2025 Form 10-K under the heading "Business—Intellectual Property—In-licensed intellectual property" and Note 12 to our consolidated financial statements contained therein.

In January 2025, we entered into a non-exclusive patent license agreement with NIAID to license certain patent rights related to the development of mRNA-based vaccines for the prevention or treatment of RSV infection. Upon execution of the agreement, we made a total payment of $10 million, which was capitalized as an intangible asset and is amortized to cost of sales on a straight-line basis over the estimated useful life of the licensed patents. In addition, we are obligated to pay low single-digit royalties on future net sales of licensed products.

For the three months ended March 31, 2026 and 2025, we recognized $895 million and $5 million, respectively, of royalty expenses, including amortization of certain intangible assets. Royalty expenses for the three months ended March 31, 2026 include $878 million related to the litigation settlement with Arbutus and Genevant. These royalty expenses were recorded to cost of sales in our condensed consolidated statements of operations.

Additionally, we have other in-license agreements with third parties which require us to make future development, regulatory and commercial milestone payments and sales-based royalties for specified products associated with the agreements. The achievement of these milestones have not yet occurred as of March 31, 2026.

13. Stock-Based Compensation and Share Repurchase Programs

Stock-Based Compensation

The following table presents the components and classification of stock-based compensation expense for the three months ended March 31, 2026 and 2025 as follows (in millions):

	Three Months Ended March 31,	
	2026	2025
Options	$ 24	$ 39
Restricted Stock Units (RSUs)	75	71
Performance Stock Units (PSUs)	3	1
Employee Stock Purchase Plan (ESPP)	2	4
Total	$ 104	$ 115
Cost of sales	$ 3	$ 7
Research and development	64	69
Selling, general and administrative	37	39
Total	$ 104	$ 115

As of March 31, 2026, there was $978 million of total unrecognized compensation cost related to unvested stock-based compensation with respect to options, RSUs and PSUs granted. That cost is expected to be recognized over a weighted-average period of 2.9 years as of March 31, 2026.

Share Repurchase Programs

As of March 31, 2026, $1.7 billion of our Board of Directors' authorization for repurchases of our common stock (the 2022 Repurchase Programs) remains outstanding, with no expiration date. There were no shares repurchased during the three months ended March 31, 2026 or 2025.

14. Income Taxes

The following table summarizes our income tax expense for the periods presented (in millions, except for percentages):

	Three Months Ended March 31,	
	2026	2025
Loss before income taxes	$ (1,334)	$ (964)
Provision for income taxes	$ 9	$ 7
Effective tax rate	(0.7)%	(0.7)%

The effective tax rate for the three months ended March 31, 2026 was higher than the statutory rate, primarily due to our global valuation allowance, which limits our ability to recognize tax benefits from the loss. The higher effective tax rate was also impacted by certain of our foreign subsidiaries that have taxable income, while we incurred a net loss before income taxes in other jurisdictions. The effective tax rate for the three months ended March 31, 2026 was consistent with the same period in 2025, primarily due to the continued maintenance of global valuation allowance. For additional details regarding our deferred tax assets and the policies governing our valuation allowance, please refer to Note 14 to our consolidated financial statements in our 2025 Form 10-K.

We periodically reassess the need for valuation allowances on our deferred tax assets, considering both positive and negative evidence to evaluate whether it is more likely than not that all or a portion of such assets will not be realized. Significant management judgment is required in assessing the realizability of our deferred tax assets. In the event that actual results differ from our estimates, we adjust our estimates in future periods and we may need to modify our valuation allowance, which could materially impact our financial position and results of operations.

We file income tax returns in the U.S. and various state, local and foreign jurisdictions. The income tax returns of all material taxing jurisdictions remain open to tax examination for all tax years since our date of incorporation in those jurisdictions. As of March 31, 2026, we are under audit in various U.S. and foreign jurisdictions. No adjustments to our tax positions have been proposed at this time.

15. Net loss per Share

The computation of basic earnings (loss) per share (EPS) is based on the weighted-average number of our common shares outstanding. The computation of diluted EPS is based on the weighted-average number of our common shares outstanding and potential dilutive common shares during the period as determined by using the treasury stock method.

Basic and diluted EPS for the three months ended March 31, 2026 and 2025 were calculated as follows (in millions, except per share data):

	Three Months Ended March 31,	
	2026	2025
Numerator:		
Net loss	$ (1,343)	$ (971)
Denominator:		
Basic and diluted weighted-average common shares outstanding	395	386
Basic and diluted EPS	$ (3.40)	$ (2.52)
Common stock equivalents excluded from the EPS computation above because their inclusion would have been anti-dilutive	45	49

2. MANAGEMENT'S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS

You should read the following discussion and analysis of our financial condition and results of operations together with our unaudited financial information and related notes included in this Form 10-Q and our consolidated financial statements and related notes and other financial information in our Annual Report on Form 10-K for the fiscal year ended December 31, 2025, which was filed with the Securities and Exchange Commission (the SEC) on February 20, 2026 (the 2025 Form 10-K).

Overview

We are a biotechnology company advancing a new class of medicines made of messenger RNA (mRNA). mRNA medicines are designed to direct the body's cells to produce intracellular, membrane or secreted proteins that have a therapeutic or preventive benefit with the potential to address a broad spectrum of diseases. Our platform builds on continuous advances in basic and applied mRNA science, delivery technology and manufacturing, providing us the capability to pursue in parallel a robust pipeline of new development candidates. We are developing medicines across infectious disease vaccines, oncology therapeutics and rare disease therapeutics.

Since our founding in 2010, we have transformed from a research-stage company advancing programs in the field of mRNA to a commercial enterprise with a diverse clinical portfolio of vaccines and therapeutics across several modalities, a broad intellectual property portfolio and integrated manufacturing capabilities that allow for rapid clinical and commercial production at scale. We currently have four approved products—Spikevax® and mNEXSPIKE®, our COVID vaccines; mRESVIA®, our vaccine against respiratory syncytial virus (RSV); and mCOMBRIAX®, our flu plus COVID combination vaccine, which was recently approved in Europe for individuals 50 years of age and older. We also have a diverse development pipeline of 25 development candidates across our 35 development programs currently in clinical studies.

Business Highlights

Strategic Collaboration with Recordati

In January 2026, we announced a strategic collaboration with Recordati S.p.A. (Recordati) to advance our investigational propionic acidemia (PA) therapeutic, mRNA-3927, through the final stages of clinical development and, upon approval, global commercialization. The transaction closed on March 16, 2026. Under the agreement, we will continue to lead clinical development of mRNA-3927 through approval, and Recordati will lead commercialization. Recordati has an established global commercial infrastructure and expertise in rare diseases, including PA, which is expected to support commercialization upon approval. mRNA-3927 targets a serious rare metabolic disease with significant unmet medical need and is currently being evaluated in a registrational study that has reached target enrollment, with potential data expected in 2026. Under the terms of the agreement, Recordati agreed to make an upfront payment of $50 million and are we are eligible to receive up to an additional $110 million in development and regulatory milestone payments, in addition to commercial and sales milestones and tiered royalties on net sales.

Strategic Agreement with the Government of Mexico

In February 2026, we signed a memorandum of understanding for a long-term strategic agreement with the Government of Mexico, Laboratorios de Biológicos y Reactivos de Mexico (BIRMEX), and Laboratorios Liomont (Liomont) to support the development of local mRNA manufacturing capabilities and strengthen pandemic preparedness. The agreement includes the supply of our respiratory vaccine portfolio and a technology transfer to Liomont to enable domestic manufacturing of our COVID vaccine, mRNA-1273. In addition, the collaboration is expected to support local clinical research and development activities aligned with Mexico's public health priorities.

Settlement with Arbutus and Genevant

On March 3, 2026, we entered into a settlement agreement with Arbutus Biopharma Corporation (Arbutus) and Genevant Sciences GmbH (Genevant, and with Arbutus, Arbutus/Genevant) resolving all litigation worldwide, including between the parties in the U.S. District Court for the District of Delaware. The settlement resolves all worldwide Arbutus/Genevant litigation related to Spikevax and mRESVIA and provides certainty going forward for our full infectious disease portfolio, including mNEXSPIKE, mCOMBRIAX and our future vaccine pipeline, with no future royalties owed. Under the terms of the agreement, we agreed to make a lump sum payment of $950 million, which is payable in the third quarter of 2026. Consistent with the terms of the settlement agreement, we have appealed the District Court's decision related to 28 U.S.C. § 1498 to the Federal Circuit Court of Appeals and could be required to make an additional payment of up to $1.3 billion depending on the outcome.

European Commission Marketing Authorization for mCOMBRIAX

In April 2026, we received marketing authorization from the European Commission (EC) for mCOMBRIAX (mRNA-1083), our mRNA combination vaccine for the prevention of influenza disease and COVID-19 in individuals 50 years of age and older. The marketing authorization follows a positive opinion from the European Medicines Agency's (EMA) Committee for Medicinal Products for Human Use (CHMP) and is valid in all 27 European Union member states, as well as Iceland, Liechtenstein and Norway. mCOMBRIAX is our fourth authorized product and further strengthens our respiratory portfolio and commitment to the European Union. The vaccine builds on advances from the clinical development of mNEXSPIKE and mRNA-1010, our investigational seasonal influenza vaccine. mCOMBRIAX will be made available across the European Union, subject to national regulatory and access procedures, and we are working with national authorities to support local access and implementation.

Total Revenue and Net Loss Per Share

For the first quarter of 2026, we recognized total revenue of $389 million, compared to $108 million for the first quarter of 2025. Net loss per share was $(3.40) for the first quarter of 2026, compared to net loss per share of $(2.52) for the first quarter of 2025.

Recent Program Developments

Infectious Disease Vaccines

- *Seasonal flu + COVID vaccine:* We received EC marketing authorization for mCOMBRIAX in the EU and our mRNA-1083 regulatory filings are under review in Canada and Australia. We are awaiting further guidance from the U.S. Food and Drug Administration (FDA) on refiling the submission for mRNA-1083. In addition, we recently presented mRNA-1083 data from a Japanese cohort at the 2026 European Society of Clinical Microbiology and Infectious Diseases (ESCMID) Global Congress.

- *Seasonal flu vaccine:* The FDA has assigned a Prescription Drug User Fee Act (PDUFA) goal date for mRNA-1010 of August 5, 2026. Our mRNA-1010 regulatory filings are also under review in Europe, Canada and Australia, and potential approvals are expected to begin in 2026. We recently presented mRNA-1010 revaccination data at the 2026 ESCMID Global Congress.

- *Norovirus vaccine:* Our ongoing Phase 3 safety and efficacy study of our trivalent vaccine candidate against norovirus (mRNA-1403) is fully enrolled in a second Northern Hemisphere season (2025-2026) with data expected in 2026, subject to case accruals.

Oncology Therapeutics

- *Intismeran autogene:* We are advancing mRNA-4157 in collaboration with Merck, with nine total Phase 2 and Phase 3 clinical trials underway across multiple tumor types, including melanoma, non-small cell lung cancer (NSCLC), bladder cancer and renal cell carcinoma. This includes the recent initiation of a Phase 3 study of intismeran as monotherapy and in combination with KEYTRUDA QLEX for the treatment of high-risk Stage 1 NSCLC.

  Fully enrolled studies include a Phase 3 adjuvant melanoma, a Phase 2 adjuvant renal cell carcinoma, and a Phase 2 adjuvant muscle invasive bladder cancer. We expect Phase 3 adjuvant melanoma data potentially in 2026.

  We recently announced an upcoming oral presentation at the 2026 American Society of Clinical Oncology (ASCO) Annual Meeting highlighting positive five-year Phase 2b adjuvant melanoma data, which showed a sustained benefit with intismeran in combination with KEYTRUDA, reducing the risk of recurrence or death by 49% compared to KEYTRUDA alone.

- *mRNA-4359.* Our Phase 1/2 study of mRNA-4359, an investigational wholly-owned cancer antigen therapy, is ongoing. We recently presented mRNA-4359 data at the American Association for Cancer Research (AACR) 2026 Annual Meeting. The Phase 2 portion of the study includes cohorts in first-line metastatic melanoma, second-line+ metastatic melanoma and first-line metastatic NSCLC, and we expect a potential Phase 2 data readout in 2026.

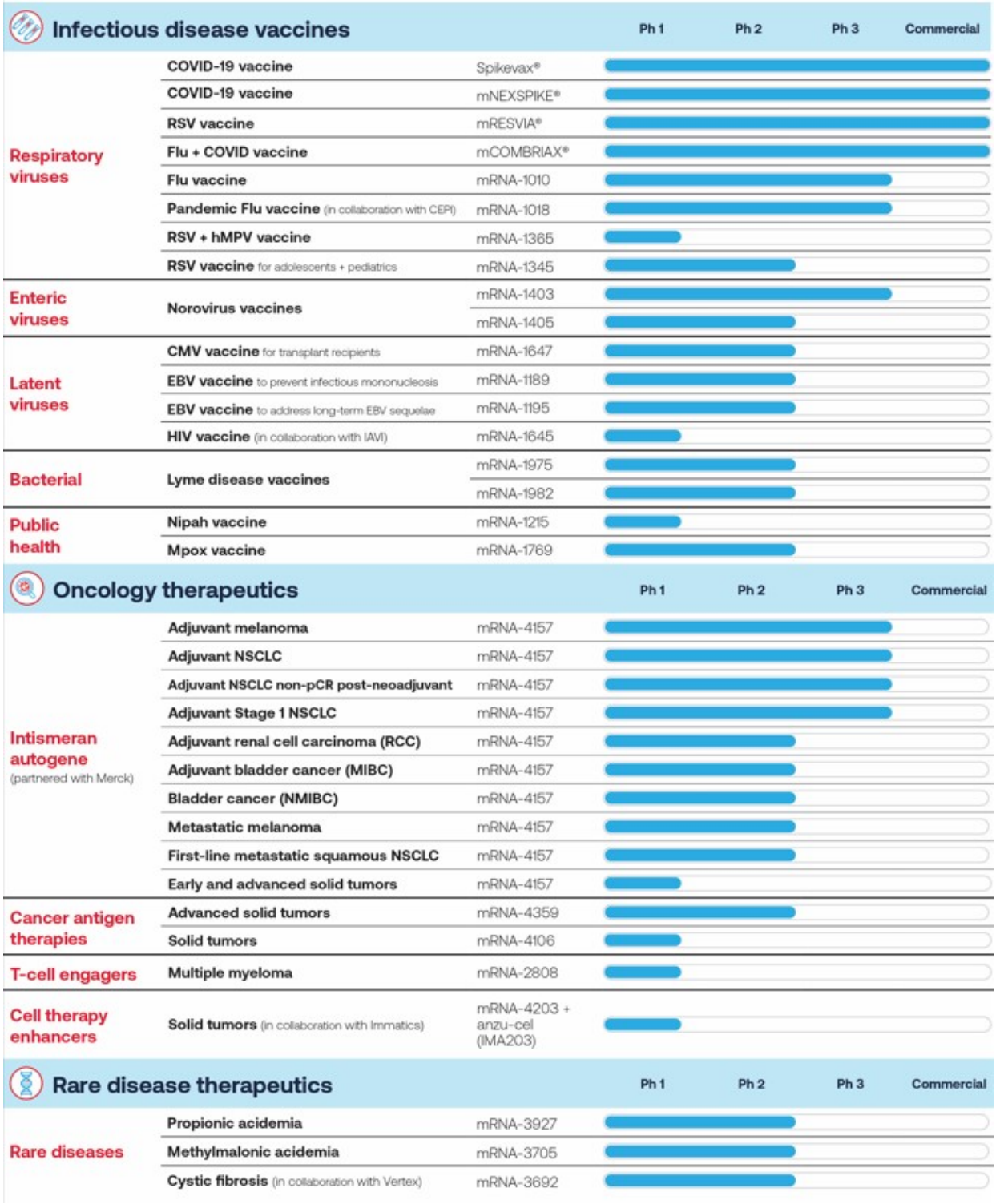
Rare Disease Therapeutics

- *Propionic acidemia (PA) therapeutic:* Our investigational therapeutic for PA (mRNA-3927) is in a registrational study and target enrollment has been reached.

- *Methylmalonic acidemia (MMA) therapeutic:* We are deferring our decision on a pivotal trial for mRNA-3705 until PA registrational data readout.

Our Pipeline

The following chart shows our current pipeline of 35 development programs across our several modalities.



Abbreviations: CMV, cytomegalovirus; EBV, Epstein-Barr virus; HIV, human immunodeficiency virus; hMPV, human metapneumovirus; MIBC, muscle invasive bladder cancer; NMIBC, non-muscle invasive bladder cancer; NSCLC, non-small cell lung cancer; pCR, pathological complete response; RCC, renal cell carcinoma; RSV, respiratory syncytial virus.

Results of operations

The following table summarizes our condensed consolidated statements of operations for the periods presented (in millions):

	Three Months Ended March 31,		Change 2026 vs. 2025	
	2026	2025	$	%
Revenue:				
Net product sales	$ 352	$ 86	$ 266	309%
Other revenue	37	22	15	68%
Total revenue	389	108	281	260%
Operating expenses:				
Cost of sales	955	90	865	961%
Research and development	649	856	(207)	(24)%
Selling, general and administrative	173	212	(39)	(18)%
Total operating expenses	1,777	1,158	619	53%
Loss from operations	(1,388)	(1,050)	(338)	32%
Interest income	72	90	(18)	(20)%
Other expense, net	(18)	(4)	(14)	350%
Loss before income taxes	(1,334)	(964)	(370)	38%
Provision for income taxes	9	7	2	29%
Net loss	$ (1,343)	$ (971)	$ (372)	38%

Revenue

Net product sales

Net product sales by customer geographic location were as follows (in millions):

	Three Months Ended March 31,	
	2026	2025
United States	$ 73	$ 31
Europe	239	—
Rest of world	40	55
Total	$ 352	$ 86

Net product sales by product were as follows (in millions):

	Three Months Ended March 31,	
	2026	2025
COVID [(1)]	$ 345	$ 84
RSV	7	2
Total	$ 352	$ 86

[(1)] Includes sales of Spikevax and mNEXSPIKE.

As of March 31, 2026, we have three commercial products, our COVID vaccines, Spikevax and mNEXSPIKE, and our RSV vaccine, mRESVIA. We launched commercial sales of mNEXSPIKE in the third quarter of 2025.

We sell our COVID vaccines, Spikevax and mNEXSPIKE, to the commercial market as well as to foreign governments and international organizations. In the U.S., our COVID and RSV vaccines are sold primarily to wholesalers and distributors, and to a lesser extent, directly to retailers and healthcare providers. Net product sales are recognized net of estimated wholesaler chargebacks, invoice discounts for prompt payments and pre-orders, provisions for sales returns and government rebates, and other related deductions.

The following table summarizes product sales provision adjustments for the periods presented (in millions):

	Three Months Ended March 31,	
	2026	2025
Gross product sales	$ 350	$ 105
Product sales provision:		
Wholesaler chargebacks, discounts and fees	11	(22)
Returns, rebates and other fees	(9)	3
Total product sales provision adjustments	$ 2	$ (19)
Net product sales	$ 352	$ 86

Certain agreements may include upfront payments for our vaccine supply, initially recorded as deferred revenue. As of March 31, 2026, we had deferred revenue of $95 million related to product sales, of which $37 million is expected to be realized in less than one year.

Other revenue

Other revenue comprises grant revenue, collaboration revenue, licensing and royalty revenue, and stand-ready manufacturing revenue.

For the three months ended March 31, 2026, total revenue increased by $281 million, or 260%, compared to the same period in 2025, primarily driven by an increase in net product sales. Net product sales increased by $266 million, or 309%, mainly due to higher COVID vaccine sales in international markets, driven by deliveries under long-term strategic partnerships with government entities. Other revenue also increased compared to the same period in 2025, reflecting higher stand-ready manufacturing revenue from facilities under these partnerships, including those that became operational later in 2025.

Product sales are expected to return to growth in 2026, supported by the full-year impact of long-term strategic partnerships with government entities.

Operating expenses

Cost of sales

Cost of sales for the three months ended March 31, 2026 was $955 million, which included third-party royalties of $895 million, inventory write-downs of $38 million, primarily related to our finished and semi-finished COVID vaccine inventory and raw materials relative to updated demand forecasts, as well as shelf-life expiration and other adjustments. Third-party royalties included $878 million related to the litigation settlement with Arbutus and Genevant, as well as amortization of the associated intangible asset. Please refer to Note 12, Commitments and Contingencies, to our condensed consolidated financial statements for additional information. Inventory write-downs in 2026 reflected a combination of factors, including changes in demand expectations, expiration, scrap and other adjustments. Please refer to Note 7 to our condensed consolidated financial statements for inventory related charges.

Cost of sales for the three months ended March 31, 2026 increased by $865 million, or 961%, compared to the same period in 2025. Cost of sales as a percentage of net product sales and stand-ready manufacturing revenue was 249% for the three months ended March 31, 2026, compared to 92% for the corresponding period in 2025. The increase was primarily driven by litigation settlement-related expenses recognized in the first quarter of 2026. Excluding these expenses, cost of sales and cost of sales as a percentage of net product sales and stand-ready manufacturing revenue decreased by 14% and 72 percentage points, respectively, compared to the same period in 2025, primarily due to lower unutilized manufacturing capacity costs, losses on firm purchase commitments and inventory write-downs, partially offset by higher sales volume.

In 2026, we anticipate that cost of sales will remain at a relatively consistent level compared to 2025, reflecting continued manufacturing productivity improvements and operational efficiencies. This expectation excludes the impact of the settlement with Arbutus and Genevant, for which we recorded a charge of $878 million in cost of sales in the first quarter of 2026 and expect additional amortization expense in the remainder of the year. Excluding this impact, to the extent net product sales increase, cost of sales as a percentage of net product sales and stand-ready revenue may decrease modestly.

Research and development expenses

Research and development expenses decreased by $207 million, or 24%, for the three months ended March 31, 2026, compared to the same period in 2025. The decrease was mainly driven by a $48 million reduction in clinical manufacturing costs, driven by higher research and development funding from Blackstone for our seasonal flu program, a $48 million decrease in consulting and outside service costs, consistent with the lower level of clinical and manufacturing activities, and a $43 million reduction in personnel-related expenses due to lower headcount. Clinical trial expenses decreased by $42 million, reflecting lower activity in our seasonal flu, CMV, Norovirus, and RSV programs due to trial wind-downs, as well as the timing of clinical trial activities, partially offset by higher costs related to postmarketing commitments for our COVID products.

We anticipate a modest reduction in research and development expenses in 2026 compared to 2025, driven by continued portfolio prioritization, disciplined cost management, and a focused approach to pipeline execution. These reductions are expected to be partially offset by certain clinical trial activities shifting from 2025 into 2026 and additional costs related to post-marketing commitments. We remain committed to advancing our pipeline and late-stage programs, including intismeran autogene and norovirus vaccine programs, while continuing to manage research and development investment levels in line with our long-term objectives.

Selling, general and administrative expenses

For the three months ended March 31, 2026, selling, general and administrative expenses decreased by $39 million, or 18%, compared to the same period in 2025. The decrease was mainly driven by a $19 million decrease in personnel-related expenses, a $9 million decrease in marketing expenses, and an $8 million broad-based reduction in consulting and outside services. The decrease in the period largely reflects continued cost discipline and efforts to streamline operations.

We expect selling, general and administrative expenses in 2026 to remain at a level relatively consistent with 2025, reflecting an efficient and scalable operating structure. While we will continue to make selective investments to support our key priorities, including our global commercial and regulatory activities, we expect these investments to be largely offset by ongoing efficiency initiatives and disciplined resource allocation.

Interest income

For the three months ended March 31, 2026, interest income decreased by $18 million, or 20%, compared to the same period in 2025. The decrease was primarily due to lower average investment balances and interest rates.

Other expense, net

The following tables summarize other expense, net for the periods presented (in millions):

	Three Months Ended March 31,		Change 2026 vs. 2025	
	2026	2025	$	%
Loss on investments	$ (1)	$ (7)	$ 6	(86)%
Interest expense	(17)	(1)	(16)	1,600%
Other income, net	—	4	(4)	(100)%
Total other expense, net	$ (18)	$ (4)	$ (14)	350%

For the three months ended March 31, 2026, total other expense, net increased by $14 million, or 350%, compared to the same period in 2025. The increase was largely driven by higher interest expense. Interest expense is primarily related to our finance leases related to certain contract manufacturing service agreements and, beginning in November 2025, interest associated with our long-term debt. Please refer to Note 10 and Note 11 to our condensed consolidated financial statements for additional information.

Income taxes

Provision for income taxes for the three months ended March 31, 2026, remained immaterial and consistent with the same periods in 2025. The effective tax rate continues to reflect the maintenance of our global valuation allowance, which limits our ability to recognize tax benefits from the losses. Please refer to Note 14 to our condensed consolidated financial statements for additional details.

Liquidity and capital resources

The following table summarizes our cash, cash equivalents, investments and working capital as of March 31, 2026 and December 31, 2025 (in millions):

	March 31, 2026	December 31, 2025
Financial assets:		
Cash and cash equivalents	$ 1,908	$ 2,595
Investments	3,297	3,204
Investments, non-current	2,251	2,336
Total	$ 7,456	$ 8,135
Working capital:		
Current assets	$ 5,770	$ 6,544
Current liabilities	2,395	1,987
Total	$ 3,375	$ 4,557

Our cash, cash equivalents and investments are invested in accordance with our investment policy, primarily with a view to liquidity and capital preservation. Investments, consisting primarily of government and corporate debt securities, are stated at fair value. Cash, cash equivalents and investments as of March 31, 2026 decreased by $679 million, or 8%, compared to December 31, 2025. The decrease in cash, cash equivalents and investments was primarily due to a net cash outflow from operating activities of $630 million and purchases of property and equipment of $62 million during the three months ended March 31, 2026.

Working capital, defined as current assets less current liabilities, decreased by $1.2 billion, or 26%, as of March 31, 2026, compared to December 31, 2025. This was primarily driven by a decrease in cash, cash equivalents and current investments of $594 million to fund operations, an increase in accrued liabilities and accounts payable of $370 million, including $950 million related to the litigation settlement, partially offset by lower spend during the period, and a decrease in accounts receivable of $113 million primarily driven by timing of collections.

As of March 31, 2026, we did not have any off-balance sheet arrangements. For a discussion of our contractual obligations and commitments, refer to our 2025 Form 10-K.

Cash flow

The following table summarizes the primary sources and uses of cash for each period presented (in millions):

	Three Months Ended March 31,	
	2026	2025
Net cash (used in) provided by:		
Operating activities	$ (630)	$ (1,037)
Investing activities	(76)	730
Financing activities	17	4

Operating activities

We derive cash flows from operations primarily from cash collected from customer advance payments and accounts receivable related to our product sales, as well as other revenue and funding arrangements. Our cash flows from operating activities are significantly affected by our use of cash for operating expenses and working capital to support the business. We sell our COVID and RSV vaccines to the commercial market as well as to foreign governments and international organizations. Certain supply agreements include upfront payments, which are initially recorded as deferred revenue. In the U.S., our COVID and RSV vaccines are sold primarily to wholesalers and distributors, and to a lesser extent, directly to retailers and healthcare providers. Wholesalers and distributors typically do not make upfront payments to us. In addition, we receive customer advance payments related to certain other revenue arrangements. As of March 31, 2026, we had $256 million in deferred revenue related to customer advance payments received or billable.

Net cash used in operating activities for the three months ended March 31, 2026 was $630 million and consisted of net loss of $1.3 billion, non-cash adjustments of $160 million and a net change in assets and liabilities of $553 million. Non-cash items primarily included stock-based compensation of $104 million, and depreciation and amortization of $59 million. The net change in assets and liabilities was mainly due to an increase in accrued liabilities and accounts payable of $344 million, including $950 million related to the litigation settlement, partially offset by lower spend during the period, a decrease in accounts receivables, net of $114 million due to timing of collections, and a decrease in prepaid and other assets of $54 million, driven by a decrease in vendor prepayments.

Net cash used in operating activities decreased by $407 million, or 39%, during the three months ended March 31, 2026, compared to the same period in 2025, primarily attributable to a change in accrued liabilities and accounts payable of $881 million, including $950 million related to the litigation settlement. This decrease was partially offset by an increase in net loss of $372 million and a decrease in accounts receivable, net of $166 million, driven by timing of collections.

Investing activities

Our primary investing activities consist of purchases, sales, and maturities of our investments, capital expenditures for facilities, manufacturing and laboratory equipment, and computer equipment and software, as well as business development activities.

Net cash used in investing activities for the three months ended March 31, 2026 was $76 million, driven primarily by purchases of marketable securities of $1.3 billion, and purchases of property and equipment of $62 million, partially offset by proceeds from maturities and sales of marketable securities of $1.3 billion.

Net investing cash flows decreased by $806 million, or 110%, during the three months ended March 31, 2026, compared to the same period in 2025, primarily due to a decrease in proceeds from maturities and sales of marketable securities of $1.3 billion, partially offset by a decrease in purchases of marketable securities of $416 million, and a decrease in purchases of plant, property and equipment of $55 million.

Financing activities

Net cash provided by financing activities for the three months ended March 31, 2026 was $17 million, primarily related to proceeds from issuance of common stock through equity plans.

Net cash provided by financing activities increased by $13 million, or 325%, during the three months ended March 31, 2026, compared to the same period in 2025, mainly due to an increase in proceeds from issuance of common stock through equity plans of $16 million.

Operation and funding requirements

Our principal sources of funding as of March 31, 2026 consisted of cash and cash equivalents, investments, and cash we may generate from operations. We reported a net loss of $1.3 billion for the three months ended March 31, 2026 and net losses of $2.8 billion and $3.6 billion for the years 2025 and 2024, respectively. Historically, from our inception to the end of 2020, we incurred significant losses from operations due to our significant research and development expenses. Following the authorization of our first commercial product in December 2020, we generated significant net income in both 2022 and 2021. We have retained earnings of $5.9 billion as of March 31, 2026.

We have significant future capital requirements including expected operating expenses to conduct research and development activities, operate our organization, and meet capital expenditure needs. We anticipate maintaining substantial expenses across all areas of our ongoing activities, particularly as we continue research and development of our development candidates and clinical activities for our investigational medicines. This also extends to our manufacturing costs, including our arrangements with our supply and manufacturing partners. Our ongoing work on our intismeran autogene, norovirus, and flu+COVID combination vaccine candidates, our other oncology programs, development of any new COVID vaccines against variants of SARS-CoV-2, late-stage clinical development, investments in digital capabilities and artificial intelligence technologies, and buildout of global commercial, regulatory, sales and marketing infrastructure and manufacturing facilities will require significant cash outflows in future periods, most of which will not be reimbursed or otherwise paid for by our collaborators or alliances. We may also incur additional costs related to postmarketing commitments, though the timing and scope of such commitments remain uncertain. In addition, we have substantial facility, lease and purchase obligations. We have entered into various collaboration and licensing agreements, as well as research and development funding arrangements with third parties. These arrangements collectively encompass the funding of specific research and development activities, with the distinction that under the research and development funding arrangement, we receive funding. However, for all these arrangements, we may be obligated to make potential future milestone and royalty payments.

In November 2025, we entered into a Credit and Guaranty Agreement (Credit Agreement) with lenders led by Ares Capital Corporation, as administrative agent, providing for a term loan facility with aggregate commitments of $1.5 billion. As of December 31, 2025, we had drawn $600 million under the initial term loan. The Credit Agreement also provides for $900 million of delayed draw term loan commitments, consisting of $400 million available, subject to applicable conditions, through November 2027 and $500 million available, subject to applicable conditions and specified regulatory approval milestones, through November 2028. Borrowings under the Credit Agreement bear interest at a variable rate based on Term SOFR or a base rate, at our option, plus an applicable margin, with interest payable periodically and the principal amount due in full at maturity in November 2030. The Credit Agreement includes customary affirmative and negative covenants, with which we were in compliance as of March 31, 2026. See Note 11 to our condensed consolidated financial statements for additional information.

We believe that our cash, cash equivalents, and investments as of March 31, 2026, together with cash expected to be generated from product sales and available borrowings under our credit facility, will be sufficient to enable us to fund our projected operations and capital expenditures through at least the next 12 months from the issuance of these financial statements included in this Form 10-Q. We are subject to all the risks related to the development and commercialization of novel medicines, and we may encounter unforeseen expenses, difficulties, complications, delays, and other unknown factors, which may adversely affect our business. For example, we experienced a decline in customer demand for our COVID vaccine in 2023 and 2024, and this trend continued in 2025 as the market transitions to a more competitive and commercially driven environment, with broader external factors continuing to affect market dynamics. We foresee that our commitment to investing in our business for future product launches may lead to continued negative cash flows from operations in upcoming periods. Our forecast of the period of time through which our financial resources will be adequate to support our operations is a forward-looking statement and involves risks and uncertainties, and actual results could vary as a result of a number of factors. We have based this estimate on assumptions that may prove to be wrong, and we could utilize our available capital resources sooner than we currently expect.

Critical accounting policies and significant judgments and estimates

There have been no material changes in our critical accounting policies and estimates in the preparation of our condensed consolidated financial statements during the three months ended March 31, 2026 compared to those disclosed in our 2025 Form 10-K.

Contractual Obligations

As of March 31, 2026, there have been no material changes to our contractual obligations and commitments from those described under “Management’s Discussion and Analysis of Financial Condition and Results of Operations” included in our 2025 Form 10-K.

Item 3. Quantitative and Qualitative Disclosures about Market Risk

Our market risks, and the way we manage them, are summarized in Part II, Item 7A., “Quantitative and Qualitative Disclosures About Market Risk” of our 2025 Form 10-K. There have been no material changes to our market risk or to our management of such risks for the three months ended March 31, 2026.

Item 4. Controls and Procedures

Disclosure Controls and Procedures

Our management, with the participation of our Chief Executive Officer and our Chief Financial Officer, evaluated the effectiveness of our disclosure controls and procedures as of March 31, 2026. The term “disclosure controls and procedures,” as defined in Rules 13a-15(e) and 15d-15(e) under the Exchange Act means controls and other procedures of a company that are designed to ensure that information required to be disclosed by a company in the reports that it files or submits under the Exchange Act is recorded, processed, summarized and reported, within the time periods specified in the SEC’s rules and forms. Disclosure controls and procedures include, without limitation, controls and procedures designed to ensure that information required to be disclosed by a company in the reports that it files or submits under the Exchange Act is accumulated and communicated to the company’s management, including its principal executive and principal financial officers, as appropriate to allow timely decisions regarding required disclosure. Management recognizes that any controls and procedures, no matter how well designed and operated, can provide only reasonable assurance of achieving their objectives and management necessarily applies its judgment in evaluating the cost-benefit relationship of possible controls and procedures. Based on the evaluation of our disclosure controls and procedures as of March 31, 2026, our Chief Executive Officer and Chief Financial Officer concluded that, as of such date, our disclosure controls and procedures were effective at the reasonable assurance level.

Changes in Internal Control over Financial Reporting

There were no changes in our internal control over financial reporting (as defined in Rules 13a-15(f) and 15d-15(f) under the Exchange Act) during the three months ended March 31, 2026 that have materially affected, or are reasonably likely to materially affect, our internal control over financial reporting.

Inherent Limitations on Effectiveness of Controls

Our management, including our Chief Executive Officer and Chief Financial Officer, believes that our disclosure controls and procedures and internal control over financial reporting are designed to provide reasonable assurance of achieving their objectives and are effective at the reasonable assurance level. However, our management does not expect that our disclosure controls and procedures or our internal control over financial reporting will prevent all errors and all fraud. A control system, no matter how well-conceived and operated, can provide only reasonable, not absolute, assurance that the objectives of the control system are met. Further, the design of a control system must reflect the fact that there are resource constraints, and the benefits of controls must be considered relative to their costs. Because of the inherent limitations in all control systems, no evaluation of controls can provide absolute assurance that all control issues and instances of fraud, if any, have been detected. These inherent limitations include the realities that judgments in decision making can be faulty, and that breakdowns can occur because of a simple error or mistake. Additionally, controls can be circumvented by the individual acts of some persons, by the collusion of two or more people or by a management override of the controls. The design of any system of controls also is based in part upon certain assumptions about the likelihood of future events, and there can be no assurance that any design will succeed in achieving its stated goals under all potential future conditions; over time, controls may become inadequate because of changes in conditions, or the degree of compliance with policies or procedures may deteriorate. Because of the inherent limitations in a cost-effective control system, misstatements due to error or fraud may occur and not be detected.

PART II

Item 1. Legal Proceedings

We are involved in various claims and legal proceedings of a nature considered ordinary course in our business, including those described in our 2025 Form 10-K under the heading “Legal Proceedings.” Most of the issues raised by these claims are highly complex and subject to substantial uncertainties. For a description of risks relating to these and other legal proceedings we face, see Part I, Item 1A., “Risk Factors,” of our 2025 Form 10-K, including the discussion under the headings entitled “Risks related to our intellectual property” and “Risks related to the manufacturing of our commercial products and product candidates.” The outcome of any such proceedings, regardless of the merits, is inherently uncertain; therefore, assessing the likelihood of loss and any estimated damages is difficult and subject to considerable judgment.

Arbutus Patent Litigation

As more fully described in our 2025 Form 10-K, Arbutus Biopharma Corporation (Arbutus) and Genevant Sciences GmbH (Genevant) brought patent infringement actions against us in various jurisdictions, including in the U.S. District Court for the District of Delaware. On March 3, 2026, we entered into a settlement agreement with Arbutus and Genevant to resolve all patent infringement litigation pending worldwide. For more information, please see “Management’s Discussion and Analysis of Financial Condition and Results of Operations—Settlement with Arbutus and Genevant.”

CureVac Patent Litigation

In April 2026, CureVac SE and CureVac Manufacturing GmbH filed a complaint against us in the U.S. District Court for the District of Delaware asserting that our manufacture and sale of Spikevax willfully infringes U.S. patents directed to RNA production methods and compositions with RNA encoding modified SARS-CoV-2 spike protein. The complaint seeks a judgment of infringement of the asserted patent and unspecified damages.

Item 1A. Risk Factors

Information regarding risk and uncertainties related to our business appears in Part I, Item 1A. “Risk Factors” of our 2025 Form 10-K. There have been no material changes from the risk factors previously disclosed in the 2025 Form 10-K.

Item 2. Unregistered Sales of Equity Securities and Use of Proceeds

Issuer Purchases of Equity Securities

On August 1, 2022, our Board of Directors authorized a share repurchase program for our common stock of up to $3.0 billion, with no expiration date. During the three months ended March 31, 2026, there were no shares repurchased. As of March 31, 2026, $1.7 billion of our Board of Directors' authorization for repurchases of our common stock remains outstanding, with no expiration date.

For details about our share repurchase programs, please refer to Note 12 to our consolidated financial statements, as set forth in our 2025 Form 10-K.

Item 5. Other Information

None of our directors or officers (as defined in Rule 16a-1(f) of the Securities Exchange Act of 1934, as amended) adopted or terminated a Rule 10b5-1 trading plan or arrangement or a non-Rule 10b5-1 trading plan or arrangement, as defined in Item 408(c) of Regulation S-K, during the fiscal quarter covered by this report.

Item 6. Exhibits

The Exhibits listed below are filed or incorporated by reference as part of this Form 10-Q.

Exhibit No.	Exhibit Index
10.1*†#	Settlement Agreement, dated March 3, 2026, by and between Arbutus Biopharma Corp., Genevant Sciences GmbH, Genevant Sciences Ltd., Moderna, Inc. and ModernaTX, Inc.
31.1*	Certification of Principal Executive Officer pursuant to Rule 13a-14(a) and Rule 15d-14(a) of the Securities Exchange Act of 1934, as adopted pursuant to Section 302 of the Sarbanes-Oxley Act of 2002
31.2*	Certification of Principal Financial Officer pursuant to Rule 13a-14(a) and Rule 15d-14(a) of the Securities Exchange Act of 1934, as adopted pursuant to Section 302 of the Sarbanes-Oxley Act of 2002
32.1+	Certification pursuant to 18 U.S.C. Section 1350, as adopted pursuant to Section 906 of the Sarbanes-Oxley Act of 2002
101.INS*	XBRL Instance Document - The instance document does not appear in the Interactive Data File because its XBRL tags are embedded within the Inline XBRL document
101.SCH*	XBRL Taxonomy Extension Schema Document
101.CAL*	XBRL Taxonomy Extension Calculation Document
101.DEF*	XBRL Taxonomy Extension Definition Linkbase Document
101.LAB*	XBRL Taxonomy Extension Label Linkbase Document
101.PRE*	XBRL Taxonomy Extension Presentation Link Document
104*	Cover Page Interactive Data File (formatted as Inline XBRL with applicable taxonomy extension information contained in Exhibits 101.)

* Filed herewith

\+ The certification furnished in Exhibit 32.1 hereto is deemed to accompany this Form 10-Q and will not be deemed "filed" for purposes of Section 18 of the Securities Exchange Act of 1934, as amended. Such certification will not be deemed to be incorporated by reference into any filings under the Securities Act of 1933, as amended, or the Securities Exchange Act of 1934, as amended, except to the extent that the Registrant specifically incorporates it by reference.

† Certain portions of this exhibit (indicated by [***]) have been omitted pursuant to Item 601(b)(10)(iv) of Regulation S-K.

\# Annexes, schedules and/or exhibits have been omitted pursuant to Item 601(a)(5) of Regulation S-K. The Company undertakes to furnish supplemental copies of any of the omitted schedules or similar attachments upon request by the SEC.

SIGNATURES

Pursuant to the requirements of the Section 13 or 15(d) of the Securities Exchange Act of 1934, the Registrant has duly caused this report to be signed on its behalf by the undersigned, thereunto duly authorized.

MODERNA, INC.

Date:
May 1, 2026

By: /s/ Stéphane Bancel

Stéphane Bancel
Chief Executive Officer and Director
(*Principal Executive Officer)*

Date:
May 1, 2026

By: /s/ James M. Mock

James M. Mock
Chief Financial Officer
(*Principal Financial Officer)*

CERTAIN CONFIDENTIAL INFORMATION CONTAINED IN THIS DOCUMENT, MARKED BY "[***]", HAS BEEN OMITTED BECAUSE IT IS BOTH (I) NOT MATERIAL AND (II) IS THE TYPE THAT THE REGISTRANT TREATS AS PRIVATE OR CONFIDENTIAL.

SETTLEMENT AGREEMENT

This Settlement Agreement ("Agreement") is entered into as of March 3, 2026 (the "Effective Date") by and between, on the one hand, Genevant Sciences GmbH ("Genevant"), a limited liability company organized under the laws of Switzerland having a principal place of business at Viaduktstrasse 8, 4051 Basel, Switzerland, and Arbutus Biopharma Corp. ("Arbutus"), a British Columbia corporation having an address at 700 West Georgia St., 25th Floor, Vancouver, British Columbia V7Y1B3 Canada, and, on the other hand, Moderna, Inc., a Delaware corporation having its principal place of business at 325 Binney Street, Cambridge, Massachusetts 02142, and ModernaTX, Inc., a Delaware corporation having its principal place of business at 325 Binney Street, Cambridge, Massachusetts 02142. For purposes of this Agreement, Genevant and Arbutus shall be referred to collectively as "Genevant/Arbutus," and Moderna, Inc. and ModernaTX, Inc. shall be referred to collectively as "Moderna." The aforementioned entities are individually referred to herein as a "Party" and collectively as the "Parties." In addition, Genevant Sciences Ltd. ("Genevant Parent") is also a signatory and a "Party" to this Agreement, but solely for purposes of [***].

WHEREAS Arbutus owns, and Genevant licenses, U.S. Patent Nos. 8,058,069, 8,492,359, 8,822,668, 9,364,435, 9,504,651, and 11,141,378 (collectively, the "Asserted US Patents");

WHEREAS Genevant/Arbutus have sued Moderna in Civil Action No. 1-22-cv-00252-JDW in the U.S. District Court for the District of Delaware (the "US Litigation"), in which Genevant/Arbutus have alleged infringement of the Asserted US Patents relating to Moderna's SPIKEVAX (COVID-19 vaccine, mRNA) and Moderna has asserted certain affirmative defenses and counterclaims;

WHEREAS Moderna has alleged that, pursuant to 28 U.S.C. § 1498, it is not the proper party to compensate Genevant/Arbutus for any patent infringement in connection with sales to the United States Government under the C-100 Contract (as defined below);

WHEREAS the Court denied-in-part and granted-in-part Moderna's motion for summary judgment and granted-in-part and denied-in-part Genevant/Arbutus' motion for summary judgment concerning the application of § 1498(a) to doses supplied under the C-100 Contract in the Court's opinion and order dated February 2, 2026 [D.I. 697 and 698];

WHEREAS Genevant Parent, the indirect parent of Genevant, is not a claimant in the US Litigation or the Ex-US Litigation, is not a direct beneficiary of any potential resolution of the US Litigation or the Ex-US Litigation, is not an intended direct beneficiary of the cash Payments to Genevant/Arbutus contemplated by this Agreement, and does not directly own or license the Genevant/Arbutus Patent Rights or the EPO Patent Rights, but is a Party to this Agreement solely for purposes of the Genevant Parent Provisions;

WHEREAS the Parties wish to stipulate to the entry of judgment in the US Litigation while preserving Moderna's ability to appeal the Court's ruling with respect to the § 1498 Disputed Matter, and this Agreement sets forth the terms and conditions regarding same;

WHEREAS the Parties or their Affiliates are also involved in various patent litigation proceedings outside the United States; and

WHEREAS, to avoid the time and expense of litigation, in recognition of the inherent risks and costs of litigation and appeal, the Parties wish to resolve and settle the US Litigation and the Ex-US Litigation (as defined below) on the terms and conditions set forth below.

NOW THEREFORE, in consideration of all the above premises and terms and conditions of this Agreement, and other good and valuable consideration, the receipt and sufficiency of which are hereby acknowledged, the Parties agree as follows:

1. Definitions

The following terms used in this Agreement shall have the meanings assigned to them in this Section 1 and shall include the singular as well as the plural.

1.1 "§ 1498 Disputed Matter" means whether 28 U.S.C. §1498 bars Genevant/Arbutus' direct infringement claims and Genevant/Arbutus' indirect infringement claims against Moderna for the Disputed Doses.

1.2 "§ 1498 Judgment" means the judgment reflected in Paragraph 3a of the proposed Consent Judgment and Order for the US Litigation contained in Exhibit 2.3.

1.3 "Affiliate" means, with respect to a Person, any Person that, now or hereafter, is directly or indirectly controlled by, under common control with or that controls such Person; provided that, notwithstanding the foregoing, solely as used in Sections 1.8, 1.11, 4.1, 4.3, and 4.4 with respect to (A) each of Genevant and Genevant Parent, "Affiliate" means solely Genevant Parent and its direct and indirect wholly owned subsidiaries and (B) Arbutus, "Affiliate" means solely Arbutus Biopharma Corporation and its direct and indirect wholly owned subsidiaries. For purposes of this definition and the definition of "Change of Control," "control" or "controlled" means direct or indirect ownership or control of at least fifty percent (50%), including ownership by trusts with substantially the same beneficial interests, of the voting rights of such Person, or the power to direct the management of such Person.

1.4 "Business Day" means a day other than Saturday, Sunday, or a United States federal or Delaware state holiday.

1.5 "C-100 Contract" means Contract No. W911QY20C0100, including modifications in effect as of the Effective Date that were disclosed to Genevant/Arbutus and Genevant/Arbutus counsel, including by production in the US Litigation, prior to the Effective Date.

1.6 "Change of Control" of Moderna means any transaction or series of related transactions in which (a) a Third Party acquires, directly or indirectly, control of Moderna, Inc. or ModernaTX, Inc. (each, a "Moderna Party"), (b) a Moderna Party merges with or into a Third Party and the holders of voting rights of such Moderna Party immediately prior to such transaction do not hold more than fifty percent (50%) of the voting rights of the surviving entity immediately after such transaction, or (c) a Moderna Party sells all or substantially all of its assets to a Third Party.

1.7 "Disputed Doses" means 493,757,200 doses that were sold to the United States Government under the C-100 Contract and which were characterized by the district court in the US Litigation (D.I. 698) as "vaccines that did not go directly to United States Government employees."

1.8 "EPO Patent Rights" means:

(a) European Patent Nos. 2279254 and 4241767, and any European patent applications from which they derive;

(b) any European patents granted on the applications identified in clause (a), whether as European bundle patents under the EPC or as Unitary Patents, together with all national validations, national parts, national designations, and any extensions or validations thereof in any EPC member state or any extension or validation state;

(c) any European patent application or European patent that claims priority, directly or indirectly, in whole or in part, to: (i) any patent or application identified in clauses (a) or (b), or (ii) any priority application to which a patent or application identified in clauses (a) or (b) claims priority, in each case to the extent controlled by Genevant,
Arbutus, or any of their respective Affiliates; and

(d) any divisionals (including all generations thereof), substitutions, reissues, reexaminations, amendments, supplementary protection certificates, pediatric extensions, patent term extensions, and any other extensions or restorations of term relating to any of the foregoing.

For purposes of this definition, "controlled" means that Genevant, Arbutus, or any of their respective Affiliates (i) owns the relevant EPO Patent Right, in whole or in part, or (ii) has the legal right to license or enforce such EPO Patent Right without the consent of any Third Party and without any obligation to make any payment to any Third Party (for clarity, excluding payments to employees while employed by such Party or any of its Affiliates).

1.9 "Ex-US Litigation" means the following proceedings:

1.9.1 Canada: Federal Court of Canada File No. T-704-25, for infringement of Canadian Patent No. 2,721,333;

1.9.2 Japan: Tokyo District Court Case No. 2025 (Wa) 70079, for infringement of Japanese Patent No. 5,475,753;

1.9.3 Switzerland: Case O2025 002, for infringement of European Patent 2279254;

1.9.4 Case UPC_CFI_191/2025 (formerly Case 10280/2025), for infringement of European Patent 2279254;

1.9.5 Case UPC_CFI_192/2025 (formerly Case 10284/2025), for infringement of European Patent 4241767;

1.9.6 Moderna's opposition proceedings before the European Patent Office concerning EP 2279254; and

1.9.7 Moderna's opposition proceedings before the European Patent Office concerning EP 4241767.

1.10 "Final Non-Appealable Judgment" means a judgment, order, or decree of a court of competent jurisdiction as to which: (a) no appeal, petition for rehearing or rehearing *en banc*, or petition for writ of certiorari is pending; and (b) the time for filing any such appeal or petition has expired.

1.11 "Genevant/Arbutus Patent Rights" means: (a) the Asserted US Patents; (b) Canadian Patent No. 2,721,333, Japanese Patent No. 5,475,753, European Patent 2279254, and European Patent 4241767 (the "Asserted Ex-US Patents" and, together with the Asserted US Patents, the "Asserted Patents"); (c) on a country-by-country basis, with respect to any country anywhere in the world, all other Patents with an effective filing date before the Effective Date controlled by Genevant, Arbutus or any of their respective Affiliates that would, in the absence of the license granted under this Agreement, be infringed by the manufacture, use, distribution, offer for sale, sale, import, or export of any Moderna Licensed Product; [***] and (e) any continuations, continuations-in-part, divisionals, re-examinations, re-issues, extensions, and foreign counterparts of any of the Patents set forth in clauses (a) through (d). [***]

1.12 "Moderna Licensed Products" means: SPIKEVAX (COVID-19 vaccine, mRNA); mNEXSPIKE (COVID-19 vaccine, mRNA); mRESVIA (Respiratory Syncytial Virus vaccine); and any other mRNA vaccine, including those in development, that includes an SM-102-based LNP formulation against an infectious disease and is or will be either (i) licensed or authorized according to a Regulatory Application filed by Moderna or (ii) sold by or on behalf of Moderna or its applicable Affiliate as Moderna's or the Affiliate's own product.

1.13 "Patents" means (i) all classes or types of patents in any country or jurisdiction, including utility patents, utility models, design patents, invention certificates, reexamination certificates, and reissue patents; and (ii) all applications for all classes and types of patents in any country or jurisdiction, including provisional applications, nonprovisional applications, continuations, divisionals, and continuations-in-part.

1.14 "Person" means an individual, trust, corporation, partnership, joint venture, limited liability company, association, unincorporated organization or other legal or governmental entity.

1.15 "Regulatory Application" means an application for Emergency Use Authorization or a Biologics License Application submitted to FDA (or other application seeking approval to market a pharmaceutical product including a foreign equivalent), and any supplements, amendments, or replacements thereto.

1.16 "Third Party" means a Person that is not a Party and is not an Affiliate of a Party.

2. Releases and Dismissals

2.1 Mutual Release. Except as set forth in Section 2.1.1, effective upon the payment in full by Moderna of the noncontingent lump sum payment in accordance with Section 3.1, each Party, on behalf of itself, its Affiliates, shareholders, directors, officers, employees, agents, representatives, assigns, predecessors, and successors, hereby fully, finally and irrevocably releases, relinquishes, acquits and discharges each other Party and its Affiliates, and each of their respective directors, officers, employees, agents, representatives, heirs, assigns, shareholders, importers, manufacturers, distributors, suppliers, service providers, collaboration partners, licensees, insurers and each of their predecessors and successors of and from any and all pending and potential claims, counterclaims, demands, all manner of actions, causes of action, suits, defenses, judgments, settlements, debts, offsets, accounts, damages, losses, liabilities, costs, expenses (including expert fees), interest, punitive damages, reasonable attorneys' fees and other damages or costs of whatever nature, whether known or unknown, pending or future, certain or contingent (collectively, "Claims"), both at law and in equity, suspected or unsuspected, accrued or unaccrued, in the case of each of the foregoing based on activities on or before the Effective Date which relate in any way to: (i) the US Litigation, including the claims asserted in the US Litigation; (ii) the Ex-US Litigation, including the claims asserted in the Ex-US Litigation; and (iii) the manufacture, use, distribution, development, offer for sale, sale, import, or export of the Moderna Licensed Products or components thereof, including any claims for infringement of any of the Genevant/Arbutus Patent Rights in any jurisdiction in which the Genevant/Arbutus Patent Rights exist. For clarity, neither the releases set forth in this Section 2.1 nor any other provision of this Agreement shall prevent any Party from bringing an action or otherwise taking steps to enforce this Agreement.

2.1.1 Limitation on Release of Claims Regarding the § 1498 Disputed Matter. The Parties acknowledge that pursuant to the stipulated judgment in the US Litigation and subject to Section 2.5, Moderna may appeal from the stipulated judgment solely with respect to the § 1498 Disputed Matter. As of the Effective Date, the release provided in Section 2.1 does not extend to the Disputed Doses. If, after the Effective Date, Moderna pays in full the contingent lump sum provided in Section 3.2 and has exhausted or forfeited any remaining appellate rights concerning the decision that triggered its obligation to pay the contingent lump sum, then the release provided in Section 2.1 shall thereupon extend to the applicable Disputed Doses for which there is a Final Non-Appealable Judgment affirming the § 1498 Judgment in its entirety or otherwise holding that § 1498(a) does not bar Genevant/Arbutus' claim against Moderna for either or both of direct infringement and indirect infringement. If a Final Non-Appealable Judgment holds that § 1498(a) bars Genevant/Arbutus' claims with respect to all or some of the Disputed Doses, the release in Section 2.1 shall extend to those Disputed Doses as to which § 1498(a) was found to bar Genevant/Arbutus' claims, regardless of whether no contingent lump sum payment, or only a partial contingent lump sum payment, was required or made under Section 3.2. The release provided in Section 2.1 (including this Section 2.1.1) (i) does not extend to the United States Government, and (ii) does not result in, and shall not be construed to result in, any impairment, waiver, release or other limitation on Genevant's or Arbutus' (or any of their respective Affiliates') right to pursue any claim against the United States Government with respect to such products. Moderna shall not take any position or make any argument that is contrary to the limitations of the release, including clauses (i) and (ii) immediately above. Moderna's compliance with its contractual obligations under the C-100 Contract to the U.S. Government, including compliance with FAR 52.227-2, does not violate any term of this Agreement.

2.2 Waiver of Right to Assert Unknown Claims. Each Party hereby expressly waives any and all provisions, rights and benefits conferred by § 1542 of the California Civil Code (which provides that "A general release does not extend to claims that the creditor or releasing party does not know or suspect to exist in his or her favor at the time of executing the release and that, if known by him or her, would have materially affected his or her settlement with the debtor or released party"), or by any law or principle of common law that is similar, comparable or equivalent to § 1542 of the California Civil Code, with respect to the Claims released in Section 2.1.

2.3 Dismissal of the US Litigation and Ex-US Litigation. Within one (1) Business Day of the Effective Date, the Parties shall cause their counsels to execute and promptly file, and shall reasonably cooperate and take other reasonable actions to effectuate, stipulated judgments and stipulations of dismissal (substantially in the forms included in Exhibit 2.3 hereto) for the respective courts or tribunals to enter judgment or dismiss with prejudice or withdraw (as the case may be) all claims in the US Litigation and Ex-US Litigation, with each Party to bear its own fees and costs.

2.4 No Patent Challenges. Moderna shall not, and shall cause its Affiliates not to, voluntarily challenge or assist any Third Party in challenging the infringement, validity, or enforceability of any of the Asserted US Patents, any U.S. Patents that claim priority to the same applications as the Asserted US Patents (the "Related Asserted US Patents"), or U.S. Patent Nos. 8,492,359, 11,298,320, and 11,318,098 (collectively with the Asserted US Patents and the Related Asserted US Patents, the "Specified Patents"), except if any Specified Patent is asserted or threatened to be asserted (i) against Moderna or any of its Affiliates, or (ii) against any Related Third Party within the scope of any license, release, or covenant granted under this Agreement.

2.4.1 Freedom to Operate. Moderna, its Affiliates, and their collaboration partners may initiate or participate in any validity challenge (including IPR, PGR, reexamination, or opposition) concerning Genevant/Arbutus Patent Rights—other than Specified Patents—involving claims that are not expressly limited to the field of use of the Moderna Licensed Products and such challenge relates to Moderna products, technology, or activities—other than a Moderna Licensed Product—Moderna shall not be required to identify or specify any particular product in connection with such challenge, and any such challenge shall not constitute an admission of validity, enforceability, or infringement of the Genevant/Arbutus Patent Rights.

2.4.2 No Admissions. Nothing in this Agreement constitutes an admission by any Party of the validity or invalidity of any Genevant/Arbutus Patent Right, of direct or indirect infringement of those patents, or of any issue relating to the US Litigation or Ex-US Litigation. Further, the consent to enter judgment in Exhibit 2.3 for the US Litigation is entered into by Moderna based on the disclosures of the asserted patents' specifications, the relevant priority dates, the Court's claim construction, and the parties' contentions and expert opinions, the specific asserted relevant prior art, the specific prior related proceedings, and the asserted claims in the patents.

2.4.3 Successors and Acquired Entities. This Section 2.4 does not apply to (i) an assignee or transferee under Section 7.6 that is not an Affiliate of a Party, or (ii) any entity acquired by Moderna or its Affiliates after the Effective Date, with respect to any challenge in which that entity was participating prior to the closing of such acquisition.

2.4.4 Compelled Activities. Nothing in this Section 2.4 prevents Moderna or its Affiliates from complying with discovery requests, testifying at depositions, serving as a witness, complying with court or, subject to Section 2.6, government directives, complying with applicable laws or regulations, or complying with contractual obligations (including Moderna's obligations under the C-100 Contract, including FAR 52.227-2).

2.5 Appeal of the § 1498 Judgment. Pursuant to the stipulated judgment to be entered in the US Litigation, Moderna will preserve its right to appeal solely with respect to the §1498 Disputed Matter. For the avoidance of doubt, no Party shall challenge on appeal the district court's finding that "vaccine doses that the Government acquired and distributed to its own employees" (quoted from D.I. No. 697, at 15) are subject to 28U.S.C. § 1498(a) and Moderna is not liable for infringement. Moderna may challenge on appeal only the district court's conclusions with respect to the § 1498 Disputed Matter. In conducting these appellate proceedings at the Federal Circuit (both first-instance and following any remand), no Party will (without the consent of all other Parties) seek any extension [***]. In proceedings before the Supreme Court, no Party shall seek extension on a cert petition, and during the Supreme Court merits stage, no Party shall seek any extensions on merits briefing without first confirming with the Court that the extensions will not affect which argument session the case will be heard. Any failure to comply with the preceding sentence will be deemed a material breach of this Agreement. If any Party moves for or requests expedited oral argument in the first-instance appeal before the Federal Circuit and any subsequent appeal before the Federal Circuit following remand or other further proceedings, all other Parties shall join or support such motion or request upon written notice thereof from the moving Party.

2.6 Moderna's Role in any Court of Federal Claims Action Concerning the C-100 Contract Doses. In any claim Genevant or Arbutus (or any of their Affiliates) files against the United States Government related to doses sold under the C-100 Contract, to the extent Moderna is served with any discovery requests or subpoenas by Genevant, Arbutus, or the U.S. Government, Moderna shall reasonably comply by providing proportional discovery pursuant to a protective order that Moderna agrees to. Notwithstanding anything herein to the contrary, the provision of discovery in response to a lawful discovery request or subpoena does not breach or otherwise violate the terms of this Agreement. Notwithstanding anything herein to the contrary, Moderna shall have the right to comply with its contractual obligations under the C-100 Contract to the U.S. Government, including compliance with FAR 52.227-2, provided that Moderna shall not provide any assistance to the United States Government in any claim against the United States Government related to doses sold under the C-100 Contract that it is not contractually obligated to provide, and compliance thereunder does not breach or otherwise violate the terms of this Agreement.

3. Payments to Genevant/Arbutus

3.1 Noncontingent Lump Sum Payment. In consideration for the release and dismissals set forth in Section 2, Moderna shall pay to Genevant, Arbutus, or their respective designees, collectively, the noncreditable, nonrefundable sum of nine hundred fifty million United States Dollars (US $950,000,000.00) in accordance with an allocation schedule to be provided to Moderna by Genevant on or before June 30, 2026. Said payment shall be made on or before July 8, 2026 [***].

3.1

3.2 Contingent Lump Sum Payment. In consideration for the release and dismissals set forth in Section 2.1.1 and without limitation of Section 3.3, Moderna shall pay to Genevant, Arbutus, or their respective designees, collectively, the noncreditable, nonrefundable contingent lump sum payment specified in one of Section 3.2.1, Section 3.2.2 or Section 3.2.3, should one be applicable.

3.2.1 Full Affirmance Upon Final Non-Appealable Judgment: If there is a Final Non-Appealable Judgment affirming the § 1498 Judgment in its entirety, or otherwise holding that § 1498(a) does not bar Genevant/Arbutus' claim against Moderna for either or both of direct infringement and indirect infringement for all of the Disputed Doses, Moderna shall pay to Genevant, Arbutus, or their respective designees, collectively, one billion three hundred million United States Dollars (US $1,300,000,000.00), within ninety (90) days after such Final Non-Appealable Judgment, plus interest calculated from the date that the Federal Circuit first issues a decision with respect to the § 1498 Disputed Matter (if such decision was not a Final Non-Appealable Judgment) until the date of payment under this Section 3.2.1, at the per annum rate of the lesser of: (i) [***] over the then-current US prime lending rate (as reported in the Wall Street Journal); or (ii) the maximum rate permitted by applicable law.

3.2.2 Mixed Outcome Payment Upon Final Non-Appealable Judgment: If Section 3.2.1 does not apply and there is a Final Non-Appealable Judgment holding that
§ 1498(a) does not bar Genevant/Arbutus' claim against Moderna for either or both of direct infringement and indirect infringement for a subset of the Disputed Doses, Moderna shall pay to Genevant, Arbutus, or their respective designees, collectively, a prorated amount of one billion three hundred million United States Dollars (US $1,300,000,000.00) that reflects the number of Disputed Doses to which § 1498(a) does not bar Genevant/Arbutus' claim against Moderna.
Payment shall be due within sixty (60) days after the Final Non-Appealable Judgment. The prorated amount shall be calculated by assigning equal value to each dose, using the following formula:

(Number of Disputed Doses to which § 1498(a) does not bar) ÷ (493,757,200) × US $1,300,000,000.00 = "Mixed Outcome Payment"

3.2.3 Payment Upon Forgone Appeal: If Moderna does not file a notice of appeal within twenty-one (21) days of the district court's entry of the stipulated § 1498 Judgment, Moderna shall pay to Genevant, Arbutus, or their respective designees, collectively, one billion three hundred million United States Dollars (US $1,300,000,000.00) within ninety (90) days after such entry. If Moderna voluntarily dismisses a pending appeal, Moderna shall pay to Genevant, Arbutus, or their respective designees, collectively, one billion three hundred million United States Dollars (US $1,300,000,000.00) within ninety (90) days of such entry of filing for voluntary dismissal. Only if the voluntary dismissal was not by agreement of the Parties, interest shall be due or payable, from the Effective Date through the payment date by Moderna under this Section 3.2.3, at the per annum rate of the lesser of: (i) [***] over the then-current US prime lending rate (as reported in the Wall Street Journal); or (ii) the maximum rate permitted by applicable law. If the voluntary dismissal is made by agreement between the parties, no interest shall be due or payable by Moderna.

3.3 Interim Payments Pending Final Resolution: Because a Federal Circuit decision may be subject to further appeal, rehearing, rehearing *en banc*, certiorari, and/or remand proceedings, any payment made by Moderna pursuant to this Section 3.3 prior to a Final Non-Appealable Judgment is an interim payment that is subject to refund or true-up as set forth below.

3.3.1 Interim Payment upon Federal Circuit Full Affirmance: If there is a decision of the Federal Circuit, by the initial panel, upon panel rehearing, or *en banc*, affirming the § 1498 Judgment in its entirety, or otherwise holding that § 1498(a) does not bar Genevant/Arbutus' claim against Moderna as to either or both of direct infringement and indirect infringement, for all of the Disputed Doses, Moderna shall pay to Genevant, Arbutus, or their respective designees, collectively, one billion three hundred million United States Dollars (US $1,300,000,000.00) (the "Interim Affirmance Payment") within ninety (90) days after such Federal Circuit decision. No interest shall be due or payable by Moderna under this Section 3.3.1.

If, following the Interim Affirmance Payment, a Final Non-Appealable Judgment affirms the § 1498 Judgment in its entirety, or otherwise holds that § 1498(a) does not bar Genevant/Arbutus' claim against Moderna, as to either or both of Genevant/Arbutus' direct infringement claims and indirect infringement claims, for all of the Disputed Doses, the Interim Affirmance Payment shall be deemed to satisfy Moderna's obligation under Section 3.2, and no further contingent lump sum payment shall be due. If, however, a Final Non-Appealable Judgment reverses the § 1498 Judgment in its entirety, Genevant/Arbutus shall refund to Moderna the full amount of the Interim Affirmance Payment, plus interest calculated from the date the Interim Affirmance Payment was made by Moderna until the date of refund at the per annum rate of the lesser of: (i) [***] over the then-current US prime lending rate (as reported in the Wall Street Journal); or (ii) the maximum rate permitted by applicable law. Such refund shall be made by wire transfer of immediately available funds in accordance with instructions to be provided in writing by Moderna, within ninety (90) days after the date on which such Final Non-Appealable Judgment is entered. If the Final Non-Appealable Judgment results in a mixed outcome such that Section 3.2.2 applies, the Parties shall reconcile the Interim Affirmance Payment against the Mixed Outcome Payment, and Genevant/Arbutus shall refund any excess, plus interest as provided above, within ninety (90) days after such Final Non-Appealable Judgment.

3.3.2 Mixed Outcome Subject to Further Proceedings – Interim Payment and Binding Arbitration to Determine Interim Payment: This Section 3.3.2 applies if: (a) the Federal Circuit issues a decision that is not a Final Non-Appealable Judgment; (b) such decision holds that § 1498(a) does not bar Genevant/Arbutus' claim against
Moderna for either or both of direct infringement and indirect infringement, as to some, but not all, of the Disputed Doses; and (c) one or more Parties seek further appellate review or additional proceedings (including remand) that could alter the

number of Disputed Doses to which § 1498(a) is a bar (a "Mixed Outcome Subject to Further Proceedings").

a. Good-Faith Negotiation: Within [***] following a Mixed Outcome Subject to Further Proceedings, the Parties shall engage in good-faith negotiations to agree upon the number of Disputed Doses subject to the initial decision and the corresponding interim payment amount, if any, that Moderna must pay pending the outcome of subsequent proceedings. If the Parties reach agreement within such [***] period, Moderna shall pay the agreed-upon amount within [***] after such agreement. Payments made under this Section a are subject to refund and additional payment under the True Up provisions of Section 3.6.

b. Binding Arbitration to Determine Interim Payment: If the Parties do not reach agreement pursuant to Section a within the [***] negotiation period, either Party may invoke binding arbitration by written notice to the other Party within [***] following the expiration of the negotiation period. The arbitration shall determine: (a) the prorated number of Disputed Doses for which Moderna is likely to prevail in obtaining a final judgment that § 1498(a) bars Genevant/Arbutus' claim against Moderna for either or both of direct infringement and indirect infringement; and
(b) the corresponding interim payment amount, if any, that Moderna must pay pending the outcome of subsequent proceedings.[***].

3.4 Binding Arbitration Upon Delay of Federal Circuit Proceedings**:**

3.4.1 Right to Invoke Arbitration: If the Federal Circuit has not issued a first-instance decision by a merits panel on the § 1498 Judgment within twenty-four (24) months of Moderna's filing of the notice of appeal, Genevant/Arbutus may invoke binding arbitration by written notice to Moderna to determine whether Moderna has engaged in conduct outside of standard appellate advocacy that contributed materially to a material delay of the Federal Circuit proceedings.

3.4.2 Standard for Determination: The arbitrator shall determine solely whether Moderna engaged in conduct outside of standard appellate advocacy that contributed materially to a material delay of the Federal Circuit proceedings. For the avoidance of doubt, the following are examples of standard appellate

advocacy that shall not qualify as Moderna engaging in conduct contributing to material delay: (i) filing briefs in accordance with the standard Federal Circuit briefing schedule; (ii) efforts to generate and coordinate the filing of amicus curiae briefs regardless of the number of such briefs; (iii) responding to court orders or inquiries in accordance with the Federal Circuit's instructions, including relating to its jurisdiction to entertain the appeal; and (iv) delays caused by the court's own scheduling, internal processes, or backlog. In determining whether Moderna engaged in conduct outside of standard appellate advocacy that contributed materially to a material delay of the Federal Circuit proceedings, the arbitrator shall subtract from the asserted period of material delay any period of delay attributable to the conduct of Genevant or Arbutus and any period of delay attributable to the court.

3.4.3 Payment Obligation Upon Finding of Material Delay: If the arbitrator determines that Moderna materially delayed the Federal Circuit proceedings and is thus responsible for the absence of first-instance decision within 24 months, Moderna shall pay to Genevant, Arbutus, or their respective designees, collectively, one billion three hundred million United States Dollars (US $1,300,000,000.00) within ninety (90) days after the arbitrator's decision. Such payment shall be subject to refund, including interest, in accordance with the provisions of Section
3.10 as if it were a contingent lump sum payment made under Section 3.2.

3.4.4 Costs of Delayed Arbitration: The non-prevailing Party shall pay the prevailing Party's reasonable attorneys' fees and costs, as well as the fees and expenses of the arbitrator, as well as a lump sum of [***] as a mutually agreed reasonable estimate of the intangible costs incurred by the prevailing Party in executive time, overhead, and opportunity costs associated with the arbitration.

3.5 Arbitration Procedures: Any arbitration under Section 3.3.2(b) or Section 3.4 shall be conducted in accordance with the arbitrator selection procedures set forth in Section 3.5.1 and the arbitration procedures and schedule set forth in Section 3.5.2. Additionally, the non-prevailing Party shall pay the prevailing Party's reasonable attorneys' fees and costs, as well as the fees and expenses of the arbitrator.

3.5.1 Arbitrator Selection: Within [***] after the notice invoking arbitration, the Parties shall mutually agree upon a single neutral arbitrator with experience in patent litigation matters. If the Parties cannot agree within such period, each of Moderna and Arbutus and Genevant together shall, within [***] thereafter, designate one candidate arbitrator, and those two candidates shall jointly select the arbitrator within [***].

3.5.2 Arbitration Procedures and Schedule:

The arbitration shall proceed on the following expedited schedule:

(a) Opening Briefs: Each Party shall submit an opening brief of no more than ten (10) pages within [***] of the arbitrator's appointment. Briefs shall be submitted simultaneously (cross-briefing).

(b) Responsive Briefs: Each Party shall submit a responsive brief of no more than ten (10) pages within [***] of the exchange of opening briefs. Responsive briefs shall be submitted simultaneously (cross-briefing).

(c) Oral Argument: The arbitrator may, in the arbitrator's sole discretion, convene oral argument. If oral argument is held, it shall occur within [***] of the submission of responsive briefs.

(d) Decision: The arbitrator shall issue a written decision within [***] of the submission of responsive briefs, or within [***] of oral argument if oral argument is held. The Parties agree not to submit, cite, or reference the arbitrator's decision to any court adjudicating any appeal or remand from any appeal of the US Litigation.

3.5.3 Standard for Arbitrator's Determination in Arbitration Pursuant to Section 3.3.2(b):

The arbitrator shall determine: (a) the Party most likely to prevail in obtaining a Final Non-Appealable Judgment as to whether § 1498(a) bars Genevant/Arbutus' claim against Moderna as to either or both of direct infringement and indirect infringement, as to the disputed portion of the Disputed Doses; and (b) the proportion of the Disputed Doses as to which such Party is likely to prevail. The arbitrator shall base this determination solely on the applicable law and precedent that would govern the Federal Circuit and any court or tribunal with jurisdiction over subsequent proceedings, the record developed in the US Litigation and appellate proceedings, and the reasoning set forth in any judicial decisions issued to date.

3.5.4 Interim Payment Obligation following Arbitration Pursuant to Section 3.3.2(b): Within sixty (60) days after the arbitrator's decision, Moderna shall pay to Genevant, Arbutus, or their respective designees, collectively, an amount calculated as follows:

(Number of Disputed Doses - Number of Disputed Doses for which the arbitrator determines Moderna is likely to prevail that § 1498(a) is a bar to Genevant/Arbutus' claim against Moderna) ÷ (493,757,200) × US $1,300,000,000.00 = interim payment

If the arbitrator determines that Moderna is likely to prevail that § 1498(a) is a bar to Genevant/Arbutus' claim against Moderna as to all Disputed Doses, no interim payment shall be due.

3.6 True-Up Upon Final Non-Appealable Judgment: Upon entry of a Final Non-Appealable Judgment resolving the § 1498 Disputed Matter:

(a) Underpayment: If the Final Non-Appealable Judgment determines that Moderna is liable for a greater number of Disputed Doses than reflected in any interim payment, Moderna shall pay the difference to Genevant, Arbutus, or their respective designees within sixty (60) days after such judgment, plus interest calculated from the date that such interim payment was made until the date of payment under this Section 3.6(a), at the per annum rate of the lesser of: (i) [***] over the then-current US prime lending rate (as reported in the Wall Street Journal); or (ii) the maximum rate permitted by applicable law

(b) Overpayment: If the Final Non-Appealable Judgment determines that Moderna is liable for fewer Disputed Doses than reflected in any interim payment (including zero doses), Genevant/Arbutus shall refund the excess amount to Moderna within sixty (60) days after such judgment, plus interest calculated from the date that such interim payment was made until the date of payment under this Section 3.6(b), at the per annum rate of the lesser of:
(i) [***] over the then-current US prime lending rate (as reported in the Wall Street Journal); or (ii) the maximum rate permitted by applicable law.

3.7 Remand Proceedings: If the § 1498 Judgment is vacated or reversed and remanded to the district court for further proceedings, the issues before the district court shall be limited to the issues that the Federal Circuit remands to the district court. On remand, the Parties shall not challenge the validity, enforceability, or claim scope of any Genevant/Arbutus Patent Right, or whether the physical composition of Disputed Doses falls within the scope of the asserted claims. However, if the Federal Circuit determines that § 1498(a) bars Genevant/Arbutus's claims for direct infringement but does not bar Genevant/Arbutus' claims for indirect infringement for any Disputed Doses, Moderna shall be permitted to fully litigate on remand the number of Disputed Doses, if any, for which Moderna is liable for indirect infringement, including whether the predicate acts of direct infringement necessary to establish indirect infringement liability exist with respect to any such Disputed Doses.

3.7.1 Payment Upon Adverse Remand Decision: If (a) the § 1498 Judgment is vacated or reversed and remanded, (b) the district court determines on remand that § 1498(a) does not bar Genevant/Arbutus' claims with respect to any portion of the Disputed Doses, then Moderna shall, within ninety (90) days of such decision, pay to Genevant, Arbutus, or their respective designees, collectively, an interim payment (the "Interim Remand Payment") equal to:

(Number of Disputed Doses to which the district court determines § 1498 does not bar claims) ÷ (493,757,200) × US $1,300,000,000.00 = Interim Remand Payment

Any Interim Remand Payment shall be subject to refund or true-up upon entry of a Final Non-Appealable Judgment in accordance with Section 3.6.

3.8 Method of Payment: [***]

3.9 [***]

3.10 Return of the Contingent Lump Sum Payment: If Genevant/Arbutus receives a contingent lump sum payment pursuant to Section 3.2 based on a finding that § 1498(a) does not bar Genevant/Arbutus' claim against Moderna as to a subset of Disputed Doses and there is subsequently a Final Non-Appealable Judgment holding that § 1498(a) does bar Genevant/Arbutus' claim against Moderna as to that subset of doses, Genevant/Arbutus shall refund to Moderna the corresponding portion of the contingent lump sum payment, plus interest calculated from the date of Moderna's payment until the date of refund at the per annum rate of the lesser of: (i) [***] over the then-current US prime lending rate (as reported in the Wall Street Journal); or (ii) the maximum rate permitted by applicable law. Repayment shall be made by wire transfer within ninety (90) days of the date such amount becomes due.

3.10.1 Genevant/Arbutus Financial Covenant Protection. Any contingent lump sum payment pursuant to Section 3.2 or any interim payment pursuant to Section 3.3, in each case that remains subject to a potential repayment obligation under this Agreement (such payment, a "Potentially Refundable Payment") shall be subject to the ongoing requirements set forth in this Section 3.10.1. Genevant and Arbutus shall, from the date that such Potentially Refundable Payment becomes due until the earlier such time as any Genevant/Arbutus repayment obligation has been fully and finally extinguished, comply with the following requirements.

(a) Financial Assurance.

(i) Genevant. Moderna shall have no obligation to make any Potentially Refundable Payment unless and until the Approved Financial Assurances have been provided to Moderna sufficiently in advance for Moderna to make a timely payment. Prior to the payment by Moderna to Genevant (or its designee) of any Potentially Refundable Payment, Genevant shall provide Moderna with one or more of the following forms of alternative financial assurance ("Approved Financial Assurances"), in an aggregate amount not less than the Potentially Refundable Payment then due to Genevant (or its designee):

(a) an irrevocable surety bond issued by a surety company rated at least "A-" by A.M. Best (or an equivalent rating by a nationally recognized rating agency);

(b) one or more irrevocable standby letters of credit issued by a United States commercial bank having total assets of not less than [***];

(c) a cash escrow deposit held by an independent, nationally recognized escrow agent pursuant to an escrow agreement in form and substance reasonably acceptable to Moderna;

(d) a segregated account over which Genevant shall grant Moderna a perfected, first-priority security interest, which shall be subject to customary restrictions reasonably acceptable to Moderna; or

(e) a Parent Guaranty in accordance with the terms of Section 3.10.1(ii) below; or

(f) such other form of financial assurance as is reasonably acceptable to Moderna.

(ii) Roivant Guaranty. In the case Genevant chooses clause (a)(i)(E) as an Approved Financial Assurance, with respect to all or any portion of the Potentially Refundable Payment due to Genevant (or its designee), Roivant Sciences Ltd., a Bermuda exempted limited company ("Roivant") must execute and deliver to Moderna an unconditional, irrevocable guaranty (the "Parent Guaranty"), in a form reasonably acceptable to Moderna, pursuant to which Roivant unconditionally and irrevocably guarantees the repayment obligations of Genevant under this Agreement with respect to the applicable portion of such Potentially Refundable Payment (the "Guaranteed Refundable Payment Amount"). Any such Parent Guaranty shall:

(1) be absolute and unconditional;

(2) remain in full force and effect until the earlier of (i) the date on which Moderna's right to repayment from Genevant of the Guaranteed Refundable Payment Amount has been fully and finally extinguished, and
(ii) the date on which Genevant (or Roivant on its behalf) has repaid to Moderna the full amount of the Guaranteed Refundable Payment Amount due from Genevant hereunder (the "Parent Guaranty Period");

(3) not be subject to any defenses, set-offs, counterclaims, or conditions;

(4) waive any right of Roivant to require Moderna to proceed first against Genevant or any collateral; and

(5) provide that Roivant shall pay all amounts due within thirty (30) days after Moderna's written demand therefor following the failure of Genevant to make any required repayment when due.

If, at any time during the Parent Guaranty Period, Roivant fails to maintain cumulative cash and cash equivalents, investments (current and non-current), and outstanding liquidity related to undrawn loan facilities (determined in accordance with generally accepted accounting principles as reported in Roivant's most recent quarterly or annual filing with the SEC or equivalent regulatory authority), equal to at least two point three (2.3) times the Guaranteed Refundable Payment Amount, then Roivant or Genevant must, within thirty (30) days after such failure, provide Moderna with one or more of the Approved Financial Assurances (other than a Parent Guaranty) with respect to Guaranteed Refundable Payment Amount.

Roivant shall provide Moderna with written notice within [***] after becoming aware that Roivant's cumulative cash and cash equivalents, investments (current and non-current), and outstanding liquidity related to undrawn loan facilities has fallen below the threshold described in this Section 3.10.1(a)(ii).

(iii) Arbutus. If, at any time after the date that a Potentially Refundable Payment becomes due to Arbutus (or its designee), until the earlier of (1) the date on which Moderna's right to repayment of any portion of the Potentially Refundable Payment from Arbutus has been fully and finally extinguished, and (2) the date on which Arbutus has repaid to Moderna the full amount due by Arbutus under this Agreement with respect to a Potentially Refundable Payment, Arbutus fails to maintain cumulative cash and cash equivalents, investments (current and non-current), and outstanding liquidity related to undrawn loan facilities (determined in accordance with generally accepted accounting principles as reported in Arbutus' most recent quarterly or annual filing with the SEC or equivalent regulatory authority after giving effect, on a *pro forma* basis, to the portion of the Potentially Refundable Payment payable to Arbutus (or its designee) in accordance with Section 3.1 and not yet paid), of at least two point three times the amount of the Potentially Refundable Payment paid to Arbutus (or its designee), Arbutus shall, within [***] after such failure, provide Moderna with one or more of Approved Financial Assurances, in an aggregate amount not less than the Potentially Refundable Payment then outstanding.

Arbutus shall provide Moderna with written notice within [***] after becoming aware that Arbutus' cumulative cash and cash equivalents, investments (current and non-current), and outstanding liquidity related to undrawn loan facilities has fallen below the threshold described in this Section 3.10.1(a)(ii).

(iv) Any Approved Financial Assurance provided pursuant to this Section 3.10.1(a) shall name Moderna or its designee as beneficiary and shall remain in effect until the earlier of (A) the date on which Moderna's applicable right to repayment has been fully and finally extinguished, or (B) the date on which Genevant (or Roivant on its behalf) or Arbutus, as applicable, has repaid to Moderna the applicable full amount due. Genevant or Arbutus, as applicable, shall provide Moderna with prompt written notice of the form of Approved Financial Assurance selected and reasonable documentary evidence thereof.

(b) Effect of Failure to Satisfy Financial Covenants. If Genevant fails to deliver financial assurance pursuant to Section 3.10.1(a)(i) when due, Moderna shall deposit the Potentially Refundable Payment into an escrow account with an independent, nationally recognized escrow agent, to be released to Genevant (or its designee) upon delivery of financial assurance pursuant to Section 3.10.1(a)(i). If Arbutus fails to deliver financial assurance pursuant to Section 3.10.1(a)(iii) when due, Moderna shall deposit the portion of the Potentially Refundable Payment due to Arbutus (or its designee) into an escrow account with an independent, nationally recognized escrow agent, to be released to Arbutus (or its designee) upon delivery of financial assurance pursuant to Section 3.10.1(a)(iii).

3.11 Moderna Financial Covenant Protection. Moderna shall, until the earlier of (i) the date on which the right of Genevant, Arbutus, or their respective designees to receive any contingent lump sum payment from Moderna has been fully and finally extinguished, and (ii) the date on which Moderna has paid to Genevant, Arbutus, or their respective designees all contingent lump sum payments due under this Agreement, comply with the obligations in this Section 3.11. If, Moderna fails to maintain cumulative cash and cash equivalents, investments (current and non-current) and outstanding liquidity related to undrawn loan facilities (determined in accordance with generally accepted accounting principles as reported in Moderna's most recent quarterly or annual filing with the SEC) of at least (x) at any time between the Effective Date and the date that any contingent payment is made by Moderna pursuant to Section 3.2, three billion United States Dollars (US $3,000,000,000.00), or (y) from the date that any contingent payment is made by Moderna to Genevant, Arbutus, or their respective designees pursuant to Section 3.2, an amount equal to three billion United States Dollars (US $3,000,000,000.00) *minus* the amount of such contingent payment that was paid by Moderna to Genevant, Arbutus, or their respective designees, Moderna shall, within thirty (30) days after such failure (the "Cure Period"), either secure its ability to satisfy the contingent lump sum payment under Section 3.2.1 by providing one or more of the following forms of financial assurance, in an aggregate amount not less than, in the case of clause (x), one billion three hundred million United States Dollars (US $1,300,000,000.00), or in the case of clause (y), one billion three hundred million United States Dollars (US $1,300,000,000.00) *minus* the amount of such contingent payment that was paid by Moderna to Genevant, Arbutus, or their respective designees:

(a) an irrevocable surety bond issued by a surety company rated at least "A-" by A.M. Best (or an equivalent rating by a nationally recognized rating agency);

(b) one or more irrevocable standby letters of credit issued by a United States lender having total assets of not less than [***];

(c) a cash escrow deposit held by an independent, nationally recognized escrow agent pursuant to an escrow agreement in form and substance reasonably acceptable to Genevant/Arbutus;

(d) a segregated account over which Moderna shall grant Genevant/Arbutus a perfected, first-priority security interest, which shall be subject to customary restrictions reasonably acceptable to Genevant/Arbutus; or

(e) such other form of financial assurance as is reasonably acceptable to Genevant/Arbutus.

Any financial assurance provided pursuant to this Section 3.11 shall name Genevant, Arbutus, and their respective designees as beneficiaries and shall remain in effect until the earlier of (i) the date on which Moderna has paid all amounts payable or potentially payable by Moderna under this Agreement, or (ii) the date on which all of Moderna's obligations to make any such payments have been fully and finally extinguished. Moderna shall provide Genevant/Arbutus with (A) prompt written notice of the form of financial assurance selected and reasonable documentary evidence thereof and (B) written notice [***] after becoming aware that Moderna's cumulative cash and cash equivalents, investments (current and non-current), and outstanding liquidity related to undrawn loan facilities has fallen below the threshold described in this Section 3.11.

If Moderna fails to deliver the financial assurance pursuant to this Section 3.11 when due, Moderna shall deposit into an escrow account with an independent, nationally recognized escrow agent, to be released to Genevant, Arbutus or their respective designees when due in accordance with this Agreement, either (x) the full amount due under Section 3.2.1 (*i.e.*, one billion three hundred million United States Dollars (US $1,300,000,000.00)), or (y) if any contingent payment was made by Moderna to Genevant, Arbutus, or their respective designees pursuant to Section 3.2, an amount equal to one billion three hundred million United States Dollars (US $1,300,000,000.00) ***minus*** the amount of such contingent payment that was paid by Moderna. The escrow funds are to be released to Moderna upon delivery of financial assurance pursuant to this Section 3.11.

3.12 Tax Matters. Any payment made hereunder shall be net of applicable withholding taxes, provided that:

3.12.1 Unless a relevant change in applicable law occurs between the date of this Agreement and the date of a particular payment to Genevant hereunder, [***].
.

3.12.2 Unless a relevant change in applicable law occurs between the date of this Agreement and the date of a particular payment to Arbutus hereunder, [***].

3.12.3 If a withholding tax applies to a payment due under this Agreement, the Parties shall cooperate reasonably to complete any procedural formalities necessary to obtain any available relief, reduction or exemption from such withholding. In addition, the paying party shall timely pay the full amount deducted or withheld to the relevant governmental authority in accordance with applicable law. As soon as practicable after any payment of such taxes to a governmental authority, the paying party shall deliver to the receiving party the original or a certified copy of a receipt issued by such governmental authority evidencing such payment, a copy of the return reporting such payment or other evidence of such payment reasonably satisfactory to the receiving party.

3.12.4 In the event that there is liability of Moderna or its Affiliates to pay, or of Genevant/Arbutus or their Affiliates to collect or remit, any sales, use, value-added, or similar taxes, or any penalties or interest related thereto (collectively, "Sales Taxes") in respect of license or other rights granted to Moderna or its Affiliates under this Agreement, Moderna shall be liable for and shall pay forthwith any such Sales Taxes related thereto, and shall indemnify and hold Genevant/Arbutus and their Affiliates harmless in respect of such Sales Taxes.

3.13 Late Payments. Any late payments under this Agreement will accrue interest at the rate of [***] per month, compounded quarterly, or the highest rate allowed by applicable law, whichever is less. [***]

3.14 Limitation of Patent Rights. Moderna acknowledges and agrees that, if any Genevant/Arbutus Patent Right is materially limited or declared invalid, unpatentable or unenforceable, Moderna remains liable in full for any amounts payable under this Agreement but not yet paid and will not be entitled to the return of any payments made.

4. License to Moderna; Covenants

4.1 License. Subject to the terms and conditions of this Agreement, including Section 3.2, each of Genevant and Arbutus, on behalf of itself and its Affiliates, hereby grants to Moderna, and Moderna hereby accepts, a fully paid-up, royalty free, irrevocable, sublicensable (solely to Affiliates and Third Party CMOs and CDMOs), nontransferable (except as expressly provided in Section 7.6), nonexclusive license, under the Genevant/Arbutus Patent Rights, to develop, have developed, make, have made, use, have used, distribute, have distributed, sell, have sold, offer for sale, have offered for sale, import, have imported, export, have exported, and otherwise dispose of Moderna Licensed Products anywhere in the world. This license fully exhausts all Genevant/Arbutus Patent Rights as to Moderna Licensed Products and the practice of any method or process in connection with Moderna Licensed Products. For clarity, each Genevant/Arbutus Patent Right (regardless of jurisdiction) shall be deemed exhausted by any sale, provision, or disposition of Moderna Licensed Products, regardless of the jurisdiction in which such Moderna Licensed Products were made or first sold, to the same extent that such Genevant/Arbutus Patent Right would have been exhausted if such Moderna Licensed Products were made and first sold in the jurisdiction in which such Genevant/Arbutus Patent Right was issued.

4.2 Change of Control. In the event of a Change of Control of a Moderna Party, the licenses, releases and other rights granted to such Moderna Party under this Agreement shall remain in effect solely with respect to the Moderna Licensed Products of such Moderna Party that were in clinical development, clinically developed or commercialized by such Moderna Party or its Affiliates immediately prior to the closing of such Change of Control, and natural evolutions of such Moderna Licensed Products. The term natural evolutions includes updated versions of such products to address new viral strains or variants. Without limiting the foregoing, the licenses, releases and other rights granted to such Moderna Party under this Agreement shall not extend to or cover any products, platforms, programs or other assets of any Third Party acquirer in such Moderna Change of Control or any Affiliates of such Third Party acquirer (other than such Moderna Party and its existing Affiliates). For the avoidance of doubt, no products, platforms, programs or other assets of a Third Party shall become Moderna Licensed Products of such Moderna Party by virtue of being combined, co-formulated, co-developed, manufactured, or commercialized with a Moderna Licensed Product following a Change of Control of such Moderna Party.

4.3 Covenants to Moderna.

4.3.1 Each of Genevant, Genevant Parent, and Arbutus, on behalf of itself and its Affiliates, covenants and agrees that it and each of its Affiliates will not assert any claim for direct or indirect (including induced or contributory) infringement against Moderna or its Affiliates at any time, under any Genevant/Arbutus Patent Right anywhere in the world based on any manufacture, use, selling, offering to sell, distributing, importing, exporting and otherwise transferring or disposing of Moderna Licensed Products, past, present or future, or the practice of any process or method of operation for any such products.

4.3.2 Each of Genevant, Genevant Parent, and Arbutus, on behalf of itself and its Affiliates, covenants and agrees that it and each of its Affiliates will not assert any claim for direct or indirect (including induced or contributory) infringement against Moderna or its Affiliates at any time, under any Specified Patent or EPO Patent Right anywhere in the world based on any manufacture, use, selling, offering to sell, distributing, importing, exporting and otherwise transferring or disposing of any product (a) that is not a Moderna Licensed Product and (b) that is or will be (i) licensed or authorized according to a Regulatory Application filed by Moderna, (ii) sold by or on behalf of Moderna or its applicable Affiliate as its own product, or (iii) developed by Moderna, past, present or future, or the practice of any process or method of operation for any such products.

4.3.3 Each of Genevant, Genevant Parent, and Arbutus, on behalf of itself and its Affiliates, covenants and agrees that they will not cite, reference, rely upon, or introduce into evidence the Consent Judgment and Order (Exhibit 2.3) entered in the US Litigation (including any stipulated findings of infringement or invalidity contained therein), or any portion thereof, in any court, tribunal, arbitration, administrative proceeding, or other forum anywhere in the world, at any time, for any purpose against Moderna or any of its Affiliates.

4.4 Covenant to Related Third Parties.

4.4.1 Each of Genevant, Genevant Parent, and Arbutus, on behalf of itself and its Affiliates, covenants and agrees that it and each of its Affiliates will not assert any claim for direct or indirect (including induced or contributory) infringement against any Moderna or Moderna Affiliate customers, suppliers, manufacturers, service providers, licensees and third parties in Moderna's or a Moderna Affiliate's development, commercialization or distribution chain or with whom Moderna has entered into a collaboration agreement or license agreement (all of whom the Parties agree to be intended third-party beneficiaries of this Agreement, and collectively, "Related Third Parties") at any time under any Genevant/Arbutus Patent Right anywhere in the world, in each case based on their development, manufacture, use, selling, offering to sell, distributing, importing, exporting and otherwise transferring or disposing of Moderna Licensed Products, past, present or future, or their practice of any process or method of operation for any such Moderna Licensed Product.

4.4.2 Each of Genevant, Genevant Parent, and Arbutus, on behalf of itself and its Affiliates, covenants and agrees that it and each of its Affiliates will not assert any claim for direct or indirect (including induced or contributory) infringement against any Related Third Parties at any time under any Specified Patent or EPO Patent Right anywhere in the world, in each case based on their development, manufacture, use, selling, offering to sell, distributing, importing, exporting and otherwise transferring or disposing of any product that (a) is not a Moderna Licensed Product and (b) that is or will be (i) licensed or authorized according to a Regulatory Application filed by Moderna or its licensee, (ii) sold by or on behalf of Moderna or its applicable Affiliate or licensee as its own product, or (iii) developed by Moderna, past, present or future, or the practice of any process or method of operation for any such products.

4.5 No Implied Rights. No rights shall arise, and nothing in this Agreement shall be construed as the grant by either Party, by implication, estoppel, exhaustion, integration, or otherwise, of any intellectual property rights other than as expressly set forth in this Agreement.

5. Warranties and Representations

5.1 Mutual Representations. Each of Genevant and Arbutus, on the one hand, and Moderna, on the other hand, represents and warrants to the other(s), as of the Effective Date, that:

5.1.1 such Party is duly organized, validly existing and in good standing under the laws of the jurisdiction of its incorporation and has full corporate power and authority to enter into this Agreement and to carry out the provisions hereof;

5.1.2 such Party has taken all corporate action necessary to authorize the execution and delivery of this Agreement and the performance of its obligations under this Agreement;

5.1.3 this Agreement has been duly executed by such Party and constitutes a valid and legally binding obligation of such Party, enforceable in accordance with its terms;

5.1.4 the execution, delivery and performance of this Agreement by such Party does not conflict with any agreement, instrument or understanding, oral or written, to which it is a party or by which it may be bound, nor violate any material law or regulation of any court, governmental body or administrative or other agency having jurisdiction over it;

5.1.5 such Party has the right to grant the licenses and releases granted hereunder, and has the right to settle the US Litigation and Ex-US Litigation;

5.1.6 such Party has been advised by its counsel of its rights and obligations under this Agreement and enters into this Agreement freely, voluntarily, and without duress;

5.1.7 such Party is not relying on any promises, inducements, or representations other than those provided herein;

5.1.8 there is no lawsuit or any other civil or administrative proceeding, or any claim or counterclaim of any kind, in any court, tribunal, government entity or agency, or dispute resolution proceeding (including arbitration and mediation) pending on the Effective Date that was commenced by or on behalf of such Party or any of its Affiliates against any other Party or any of its Affiliates other than the US Litigation and the Ex-US Litigation; and

5.1.9 such Party has not transferred, assigned, subrogated or pledged to any Third Party or to an Affiliate, the right to bring, pursue, or settle any of the claims, counterclaims, or demands released pursuant to Section 2 of this Agreement.

5.2 Genevant/Arbutus Representations. Each of Genevant, Genevant Parent and Arbutus represents and warrants to Moderna, as of the Effective Date, that: (i) in the twelve (12) months prior to and including the Effective Date, neither it nor any of its Affiliates has exclusively licensed (in part or whole), transferred or assigned or purported to transfer or assign to any Person any Patent that if owned or controlled by Genevant, Arbutus, or any of their Affiliates as of the Effective Date, would be a Genevant/Arbutus Patent Right; (ii) neither it nor any of its Affiliates have participated in any way (directly or indirectly) in any transaction the purpose or effect of which is to avoid or prevent Moderna or any Related Third Parties from receiving or enjoying any part of the benefit of any of the rights, licenses, covenants, or releases provided for in this Agreement, and (iii) it has the full right and authority to enter into this Agreement on behalf of and bind all of its Affiliates, and covenants that it shall cause all such Affiliates to comply with, and grant all necessary rights, licenses, covenants, and releases to effect, this Agreement, and all other terms and conditions of this Agreement, and shall be directly liable to Moderna and its Affiliates for any breach of this Agreement by any Affiliates of Genevant or Arbutus, including any failure by any such Affiliate to grant any such right, license, covenant, or release. Each of Genevant, Genevant Parent, and Arbutus shall defend, indemnify and hold harmless Moderna and its Affiliates, and their Related Third Parties for any and all costs and expenses (including reasonable fees of attorneys and other professionals), liabilities,

damages, and losses arising out of or resulting from breach of this Section 5.2, including from any actual or threatened assertion of any Patent that would have been a Genevant/Arbutus Patent Right if the representations and warranties in this Section 5.2 were true. Genevant Parent further represents and warrants to Moderna that as of the Effective Date, it does not directly own or directly control the Genevant/Arbutus Patent Rights or the EPO Patent Rights.

5.3 No Other Warranties. Except for the express warranties and representations set forth in Section 5.1 and Section 5.2, none of the Parties makes any warranties or representations, express or implied, either in fact or by operation of law, by statute or otherwise, and each Party disclaims all other warranties and representations, including any implied warranty of fitness for a particular purpose, validity or non-infringement.

6. Confidentiality

Limitations on Disclosure. The Parties shall not, and shall cause their Affiliates and their respective employees, officers, directors and other representatives (including accountants, attorneys and auditors) not to, disclose the terms of this Agreement to any Third Party, except as provided in Sections 6.1.1 through 6.1.5.

6.1.1 The terms of this Agreement may be disclosed, without notice to the other Parties, to (i) any Party's and its Affiliates' legal counselors, auditors, accounting, financial advisors, or other similar professionals representing a Party, so long as any such Persons are bound by confidentiality obligations that extend to the terms of this Agreement; and (ii) any Third Party that is a bona fide actual or potential investor, acquirer, merger partner, or other financial partner of a Party or an Affiliate for the purpose of evaluating an investment, acquisition, merger, or similar transaction, so long as any such Third Party agrees to be bound by confidentiality requirements at least as strict as those set forth in this Agreement.

6.1.2 A Party or its Affiliate may disclose the existence and terms of this Agreement (i) as required by any applicable law, regulation, court order, legal process, stock exchange rule, or governmental authority, including the Securities Act of 1933, the Securities Exchange Act of 1934, and the rules and regulations promulgated thereunder, or (ii) in connection with any filing required to be made with the U.S. Securities and Exchange Commission ("SEC") or other applicable securities regulator. If a Party or its Affiliate determines, upon advice of counsel (internal counsel being sufficient), that it is required to file this Agreement or a description hereof with the SEC or other securities authority (including as an exhibit to a periodic report or current report), such Party or its Affiliate may do so. To the extent reasonably practicable and permitted by law, the filing Party or its Affiliate shall provide the other Party with a reasonable opportunity to review and comment on any proposed filing that describes this Agreement, and shall consider in good faith any reasonable requests for redaction of confidential information consistent with applicable securities laws and SEC rules.

6.1.3 Any Party may issue press releases or make public announcements regarding this Agreement that consist only of (i) the amount and conditions of the payments to

be made by Moderna hereunder, including the protective provisions associated therewith, (ii) any judgment or order entered into by any court related to the US Litigation or Ex-US Litigation, (iii) the general structure of the Agreement (e.g., the fact that it includes a license, releases, etc.) and (iv) content that is not a term of this Agreement. Such announcements and press releases may only be made on or after 4:15 p.m. New York time on March 3, 2026.

6.1.4 If a Party or an Affiliate is required to provide the terms of this Agreement to a Third Party pursuant to a discovery demand, discovery order, or other legal processes or requirements, including as necessary purposes of enforcing the terms of this Agreement, it shall inform the other Parties in sufficient time prior to any such disclosure to allow the other Parties to seek a protective order or confidential treatment prior to any such disclosure.

6.1.5 The terms of this Agreement may be disclosed as otherwise agreed to by the Parties in writing.

7. General Provisions

7.1 Settlement: Without limitation of the consent judgment in the US Litigation, this Agreement, including the Contingent and Non-Contingent Payments, evidences a compromise and settlement of disputed claims and nothing in this Agreement, or the fact of the Parties' respective execution of it, may be deemed or construed to be an admission of fault of any kind, which fault is expressly denied.

7.2 Term of License. The licenses granted under this Agreement shall commence upon the Effective Date and shall continue until the expiration of the last-to-expire Patent that is or becomes a Genevant/Arbutus Patent Right.

7.3 Choice of Law and Venue. This Agreement shall be governed and interpreted in accordance with the laws of the State of Delaware without regard to conflicts of law principles. The federal and state courts of Delaware shall have exclusive jurisdiction (to the extent that they have subject matter jurisdiction) in all matters arising under this Agreement, and the Parties hereto expressly consent and submit to the personal jurisdiction and venue of such courts for the limited purpose of enforcing the terms of this Agreement.

7.4 Stipulated Remedies for Certain Material Breaches. Each Party acknowledges and agrees that the restrictions and other terms and conditions set forth herein are reasonable and necessary to protect the respective legitimate interests of the Parties. The Parties also agree that irreparable damage, for which monetary damages (even if available) would not be an adequate remedy, would occur in the event of a breach of Section 2.4 or Section 2.5. Accordingly, the Parties acknowledge and agree that each of Genevant and Arbutus shall be entitled to an injunction, specific performance and other equitable relief to prevent a breach or threatened breach of Section 2.4 or Section 2.5 and to enforce specifically the terms and provisions thereof, in addition to any other remedy to which they are entitled in law or in equity. Moderna agrees that it

will not oppose the granting of such injunction, specific performance and other equitable relief on the basis that Genevant/Arbutus has an adequate remedy at law or that any award of specific performance is not an appropriate remedy for any reason at law or in equity. If either Genevant or Arbutus seeks an injunction or injunctions to prevent a breach or threatened breach of Section 2.4 or Section 2.5 and to enforce specifically such section, it shall not be required to provide any bond or other security in connection with such order or injunction.

7.5 Waiver. A waiver by any Party of any of the terms and conditions of this Agreement in any instance shall not be deemed or construed to be a waiver of such term or condition for the future, or of any subsequent breach hereof. All rights, remedies, undertakings, obligations and agreements contained in this Agreement shall be cumulative and none of them shall be in limitation of any other remedy, right, undertaking, obligation or agreement of any Party.

7.6 Bankruptcy. All rights and licenses granted under or pursuant to any Section of this Agreement are, and shall otherwise be deemed to be, for purposes of Section 365(n) of the U.S. Bankruptcy Code (the "Bankruptcy Code"), licenses of "intellectual property" as defined under the Bankruptcy Code. The Parties shall retain and may fully exercise all of their respective rights and elections under the Bankruptcy Code.

7.7 Assignment. This Agreement and the rights herein shall not be assignable or otherwise transferable without the written consent of the Parties except as provided in this Section 7.7. The prior written consent of the Parties shall not be required (i) for a Party to assign or transfer this Agreement in its entirety pursuant to a sale of all or substantially all of the assets of the entity or business division to which this Agreement relates, whether pursuant to a merger, consolidation, or similar transaction involving such Party, or (ii) for a Party to assign or transfer this Agreement in its entirety to an Affiliate; provided, however, that Moderna will in any event remain responsible for the payments to Genevant, Arbutus, or their respective designees required by Section 3 to the extent that an assignee of Moderna fails to make any such payment when due. The Agreement, as well as the rights and obligations herein, shall be binding upon and inure to the benefit of the Parties and their respective successors and assigns as if such successors and assigns were the original Parties to this Agreement or the respective rights and obligations. For avoidance of any doubt, any successors or assigns to this Agreement are permitted to assign the Agreement as if they were the original Parties to it, subject to the conditions of this Section 7.7. Any purported assignment or transfer in violation of this Section shall be null and void *ab initio*.

7.8 Assignment of Genevant/Arbutus Patent Rights. Genevant/Arbutus agrees, each on behalf of itself and each of its Affiliates, that all of the licenses, releases, covenants, and other rights granted by them and all their obligations set forth in this Agreement (the "Patent Obligations") shall run with the Genevant/Arbutus Patent Rights, and that Genevant/Arbutus shall ensure that any assignee, transferee, or successor to any of the Genevant/Arbutus Patent Rights (including the acquiring or surviving entity in connection with any acquisition or other change of control of Genevant/Arbutus), or any other Person (such as an exclusive licensee) that obtains any enforcement rights

with respect to any of the Genevant/Arbutus Patent Rights agrees in writing, prior to such assignment, transfer or grant, to be bound by the relevant Patent Obligations (including the obligation to obtain such written agreement from any subsequent assignee, transferee, successor or grantee). Genevant/Arbutus shall indemnify and hold harmless Moderna and its Affiliates, and the Related Third Parties for any costs and expenses (including reasonable fees of attorneys and other professionals), liabilities, damages and losses arising out of or resulting from any breach of this Section 7.8 by Genevant/Arbutus or their Affiliates, including from any assertion of any Genevant/Arbutus Patent Rights that would have been prevented by compliance with this Section 7.8.

7.9 Costs. Each Party shall each bear its own costs and legal fees associated with the Litigation and with the negotiation and preparation of this Agreement.

7.10 Severability. When possible, each provision of this Agreement will be interpreted in such manner as to be effective and valid under applicable law, but if any provision this Agreement is held to be prohibited by or invalid under applicable law, such provision will be ineffective only to the extent of such prohibition or invalidity, without invalidating the remainder of this Agreement.

7.11 Integration. The Agreement and its Exhibits constitute the entire agreement between the Parties with respect to the subject matter hereof and supersede all prior agreements and understandings, both oral and written, between the Parties with respect to such subject matter. No Party is relying on any promises, representations, conditions, provisions, or terms other than those set forth in this Agreement.

7.12 Amendments. No amendment, modification or supplement of any provisions of this Agreement shall be valid or effective unless made in writing and signed by a duly authorized officer of each Party.

7.13 Headings. The captions and descriptive headings of this Agreement are for convenience only, and shall be of no force or effect in construing or interpreting any of the provisions of this Agreement.

7.14 Waiver of Rule of Construction. Each Party has had the opportunity to consult with counsel in connection with the review, drafting and negotiation of this Agreement. Accordingly, any rule of construction that any ambiguity in this Agreement shall be construed against the drafting Party shall not apply.

7.15 No Third Party Beneficiaries. Except as expressly provided herein, nothing in this Agreement, either express or implied, is intended to or shall confer upon any Third Party any legal or equitable right, benefit or remedy of any nature whatsoever under or by reason of this Agreement.

7.16 Notices. Notices required or permitted under this Agreement shall be in writing and sent by prepaid registered or certified air mail or by overnight express mail

(e.g., FedEx) or by email (provided that no "bounce-back" or similar notice of non-delivery is received), and shall be deemed to have been properly served to the addressee upon receipt of such written communication to the following addresses of the Parties:

If to Moderna:
325 Binney Street,
Cambridge, MA 02142
Attention: Shannon Thyme Klinger [***]

With a copy to:
Jeanna Wacker, Kirkland & Ellis
LLP, 601 Lexington Avenue,
New York, NY 10022
[***]

If to Genevant:
Viaduktstrasse 8,
4051 Basel Switzerland
Attention:[***]
Attention: [***]

With a copy to:
David Berl [***]
Williams & Connolly LLP 680 Maine
Avenue SW Washington, D.C. 20024

If to Arbutus:
700 West Georgia St. 25th Floor
Vancouver, British Columbia V7Y1B3 Canada
Attention: [***]

With a copy to:
Adam Brausa [***]
Morrison & Foerster LLP
425 Market Street

San Francisco, California 94105

7.17 Counterparts. This Agreement may be executed in any number of signature page counterparts transmitted via facsimile or electronic mail, any one of which need not contain the signature of more than one Party but all such counterparts taken together shall constitute one and the same Agreement.

7.18 Interpretation. The words "include," "includes" and "including" shall be deemed to be followed by the phrase "without limitation." The words "herein," "hereof," and "hereunder" and other words of similar import refer to this Agreement as a whole and not to any particular Section or other subdivision. All references herein to Articles, Sections, and Exhibits shall be deemed references to Articles and Sections of, and Exhibits to, this Agreement unless the context shall otherwise require. Except where the context otherwise requires, wherever used, the singular shall include the plural, the plural the singular, the use of any gender shall be applicable to all genders and the word "or" is used in the inclusive sense (and/or). Whenever this Agreement refers to a number of days, unless otherwise specified, such number refers to calendar days.

IN WITNESS WHEREOF, each of the Parties has approved and executed this Agreement by and through its duly authorized officer or agent.

[SIGNATURE PAGE FOLLOWS]

IN WITNESS WHEREOF, the Parties have executed this Agreement effective as of the Effective Date.

Genevant Sciences GmbH **Moderna, Inc.**

By: [***]
Name: [***]
Title: [***]

By: [***]
Name: [***]
Title: [***]

Arbutus Biopharma Corp. **ModernaTX, Inc.**

By: [***]
Name: [***]
Title: [***]

By: [***]
Name: [***]
Title: [***]

Solely for purposes of the Genevant Parent Provisions

Genevant Sciences Ltd.

By: [***]
Name: [***]
Title: [***]

Exhibit 31.1

CERTIFICATION PURSUANT TO RULES 13a-14(a) AND 15d-14(a) UNDER THE SECURITIES EXCHANGE ACT OF 1934, AS ADOPTED PURSUANT TO SECTION 302 OF THE SARBANES-OXLEY ACT OF 2002

CERTIFICATIONS

I, Stéphane Bancel, certify that:

1. I have reviewed this Quarterly Report on Form 10-Q of Moderna, Inc.;

2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;

3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;

4. The registrant's other certifying officer and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) for the registrant and have:

   (a) Designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;

   (b) Designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;

   (c) Evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and

   (d) Disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and

5. The registrant's other certifying officer and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of the registrant's board of directors (or persons performing the equivalent functions):

   (a) All significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and

   (b) Any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

Date: May 1, 2026

By: /s/ Stéphane Bancel
Stéphane Bancel
Chief Executive Officer
(Principal Executive Officer)

Exhibit 31.2

CERTIFICATION PURSUANT TO RULES 13a-14(a) AND 15d-14(a) UNDER THE SECURITIES EXCHANGE ACT OF 1934, AS ADOPTED PURSUANT TO SECTION 302 OF THE SARBANES-OXLEY ACT OF 2002

CERTIFICATIONS

I, James M. Mock, certify that:

1. I have reviewed this Quarterly Report on Form 10-Q of Moderna, Inc.;

2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;

3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;

4. The registrant's other certifying officer and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) for the registrant and have:

   (a) Designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;

   (b) Designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;

   (c) Evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and

   (d) Disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and

5. The registrant's other certifying officer and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of the registrant's board of directors (or persons performing the equivalent functions):

   (a) All significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and

   (b) Any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

Date: May 1, 2026

By: /s/ James M. Mock

James M. Mock
Chief Financial Officer
(Principal Financial Officer)

Exhibit 32.1

CERTIFICATION PURSUANT TO
18 U.S.C. SECTION 1350
AS ADOPTED PURSUANT TO SECTION 906
OF THE SARBANES-OXLEY ACT OF 2002

In connection with the Quarterly Report on Form 10-Q of Moderna, Inc. (the "Company") for the period ended March 31, 2026 as filed with the Securities and Exchange Commission on the date hereof (the "Report"), we, Stéphane Bancel, Chief Executive Officer of the Company, and James M. Mock, Chief Financial Officer of the Company, certify, pursuant to 18 U.S.C. Section 1350, as adopted pursuant to Section 906 of the Sarbanes-Oxley Act of 2002, that to the best of our knowledge:

1. The Report fully complies with the requirements of Section 13(a) or 15(d) of the Securities Exchange Act of 1934, as amended; and
2. The information contained in the Report fairly presents, in all material respects, the financial condition and results of operations of the Company.

Date: May 1, 2026

By: /s/ Stéphane Bancel
Stéphane Bancel
Chief Executive Officer
(Principal Executive Officer)

Date: May 1, 2026

By: /s/ James M. Mock
James M. Mock
Chief Financial Officer
(Principal Financial Officer)